Lecture 2: Analytic Geometry

Math for Machine Learning

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Introduction

In Lecture 1, we studied vector spaces and linear transformations. Vector spaces let us add vectors and multiply them by scalars, but they do not tell us how long a vector is, how far apart two vectors are, or what the angle between two vectors is. These geometric notions are essential in machine learning:

- data points are vectors, and many algorithms, such as k-nearest neighbors and clustering, rely on distances between them; graphs of similar data points lead to spectral clustering;
- similarity between data points, such as text documents or word embeddings, is often measured by the angle between vectors;
- linear regression and principal component analysis find the orthogonal projection of data onto a subspace, and the singular values of the data matrix decide how accurately this can be computed;
- regularization penalizes the size, i.e., the norm, of a weight vector.

In this lecture, we add geometry to vector spaces, using norms and inner products.

Throughout this lecture, all vector spaces are real vector spaces, i.e., vector spaces over $\mathbb{R}$. As in Lecture 1, vectors of $\mathbb{R}^n$ are columns written in bold, $\mathbf{x} = (x_1, \ldots, x_n)^\top$, while elements of general vector spaces, such as functions, are written in normal font. We identify a 1×1 matrix with its only entry. In particular, for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

1 Norms

The length of a vector is measured by a norm. There are many reasonable ways to measure length, so we first collect the properties that every notion of length should have.

Definition 1.1 (Norm)**.** A norm on a vector space V is a function $\|\cdot\| : V \to \mathbb{R}$, $v \mapsto \|v\|$, such that for all $u, v \in V$ and all $\lambda \in \mathbb{R}$:

1. Positive definiteness: $\|v\| \geq 0$, and $\|v\| = 0$ if and only if $v = 0$;
2. Absolute homogeneity: $\|\lambda v\| = |\lambda| \, \|v\|$;
3. Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$.

A vector space together with a norm is called a normed vector space, or normed space.

The triangle inequality says that in a triangle with sides u, v and $u + v$, the length of one side is at most the sum of the lengths of the other two sides.

Definition 1.2 (ℓ_1, ℓ_2 and ℓ_∞ norms)**.** For $\mathbf{x} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$, we define

$$\begin{aligned}
\|\mathbf{x}\|_1 &= |x_1| + |x_2| + \cdots + |x_n| && \text{(the } \ell_1 \text{ norm, or Manhattan norm),} \\
\|\mathbf{x}\|_2 &= \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} = \sqrt{\mathbf{x}^\top \mathbf{x}} && \text{(the } \ell_2 \text{ norm, or Euclidean norm),} \\
\|\mathbf{x}\|_\infty &= \max\{|x_1|, |x_2|, \ldots, |x_n|\} && \text{(the } \ell_\infty \text{ norm, or maximum norm).}
\end{aligned}$$

More generally, for every real number $p \geq 1$, the ℓ_p norm is $\|\mathbf{x}\|_p = \left(|x_1|^p + \cdots + |x_n|^p\right)^{1/p}$.

The name "Manhattan norm" comes from the street grid of Manhattan: to get from $\mathbf{0}$ to $\mathbf{x}$ in $\mathbb{R}^2$ while only moving parallel to the axes, you have to walk the distance $|x_1| + |x_2|$. The Euclidean norm is the usual length of a vector, by the theorem of Pythagoras.

Example 1.3. For $\mathbf{x} = (3, -4, 1)^\top$, we have

$$\|\mathbf{x}\|_1 = 3 + 4 + 1 = 8, \qquad \|\mathbf{x}\|_2 = \sqrt{9 + 16 + 1} = \sqrt{26} \approx 5.10, \qquad \|\mathbf{x}\|_\infty = \max\{3, 4, 1\} = 4.$$

Proposition 1.4. *The ℓ_1 norm and the ℓ_∞ norm are norms on $\mathbb{R}^n$.*

Proof. Positive definiteness: both $\|\mathbf{x}\|_1$ and $\|\mathbf{x}\|_\infty$ are nonnegative, and they are 0 if and only if all $|x_i|$ are 0, i.e., $\mathbf{x} = \mathbf{0}$. Absolute homogeneity: $|\lambda x_i| = |\lambda||x_i|$, so $\|\lambda \mathbf{x}\|_1 = |\lambda| \|\mathbf{x}\|_1$ and $\|\lambda \mathbf{x}\|_\infty = |\lambda| \|\mathbf{x}\|_\infty$. Triangle inequality: for every i, $|x_i + y_i| \leq |x_i| + |y_i|$. Summing over i gives $\|\mathbf{x} + \mathbf{y}\|_1 \leq \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$, and since $|x_i| + |y_i| \leq \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$ for every i, also $\|\mathbf{x} + \mathbf{y}\|_\infty \leq \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$. □

The Euclidean norm is also a norm. Positive definiteness and absolute homogeneity are again easy to check, but the triangle inequality is harder; we prove it in Section 3 (Corollary 3.4).

A good way to visualize a norm is through its unit ball $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$. Figure 1 shows the unit balls of the three norms in $\mathbb{R}^2$.

Proposition 1.5. *For every $\mathbf{x} \in \mathbb{R}^n$,*

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq n \|\mathbf{x}\|_\infty.$$

Proof. Let $|x_k| = \|\mathbf{x}\|_\infty$ be the largest absolute value of an entry of $\mathbf{x}$. Then $\|\mathbf{x}\|_\infty^2 = x_k^2 \leq x_1^2 + \cdots + x_n^2 = \|\mathbf{x}\|_2^2$. Next,

$$\|\mathbf{x}\|_1^2 = \Big(\sum_i |x_i|\Big)^2 = \sum_i x_i^2 + \sum_{i \neq j} |x_i||x_j| \geq \sum_i x_i^2 = \|\mathbf{x}\|_2^2.$$

Finally, $\|\mathbf{x}\|_1 = \sum_i |x_i| \leq \sum_i |x_k| = n \|\mathbf{x}\|_\infty$. □

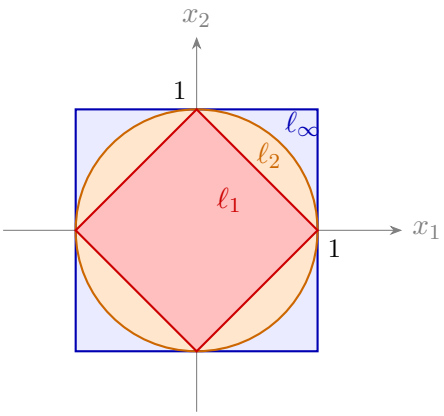


Figure 1: The unit balls $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$ of the ℓ_1 norm (red diamond), the ℓ_2 norm (orange disk) and the ℓ_∞ norm (blue square). Each ball contains the previous one, in accordance with Proposition 1.5.

So all three norms are "comparable": if a vector is small in one of them, it is also small in the others (up to a factor that depends on n). In Example 3.5 we improve the last inequality.

Example 1.6 (Frobenius norm)**.** In Lecture 1, we saw that the space $\mathbb{R}^{m\times n}$ of $m \times n$ matrices is a vector space, and it is isomorphic to $\mathbb{R}^{mn}$ by listing the entries of a matrix. The Euclidean norm of this list of entries is the Frobenius norm

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}^2} = \sqrt{\operatorname{tr}(A^\top A)},$$

which is a norm on $\mathbb{R}^{m\times n}$.

Remark 1.7 (Norms in machine learning). Norms are used to measure the size of model parameters. In regularized linear regression, we minimize the error of a model plus a penalty $\lambda\|\mathbf{w}\|$ on its weight vector $\mathbf{w}$. With the squared ℓ_2 norm (ridge regression), all weights are shrunk towards 0. With the ℓ_1 norm (the lasso), many weights become exactly 0, which gives sparse models; one reason is that the ℓ_1 unit ball in Figure 1 has corners on the coordinate axes. The ℓ_∞ norm appears, for example, when we bound how much an adversary may change each pixel of an image.

2 Inner Products

Norms measure lengths. To also measure angles, we need more structure: an inner product. The prototype is the dot product $\mathbf{x}^\top\mathbf{y}$ on $\mathbb{R}^n$.

Definition 2.1 (Inner product)**.** Let V be a vector space. A function $\langle\cdot,\cdot\rangle : V \times V \to \mathbb{R}$ is called

- bilinear if it is linear in each argument: for all $x, y, z \in V$ and $\lambda, \mu \in \mathbb{R}$,

$$\langle \lambda x + \mu y, z\rangle = \lambda\langle x, z\rangle + \mu\langle y, z\rangle \qquad \text{and} \qquad \langle z, \lambda x + \mu y\rangle = \lambda\langle z, x\rangle + \mu\langle z, y\rangle;$$

- symmetric if $\langle x, y\rangle = \langle y, x\rangle$ for all $x, y \in V$;
- positive definite if $\langle x, x\rangle > 0$ for all $x \in V$ with $x \neq 0$.

An inner product on V is a positive definite, symmetric, bilinear function $V \times V \to \mathbb{R}$. A vector space together with an inner product is called an inner product space.

Note that $\langle 0, x\rangle = \langle 0 \cdot 0, x\rangle = 0 \cdot \langle 0, x\rangle = 0$ by bilinearity, so in particular $\langle 0, 0\rangle = 0$.

Example 2.2 (Dot product)**.** The dot product $\langle \mathbf{x}, \mathbf{y}\rangle = \mathbf{x}^\top \mathbf{y} = \sum_i x_i y_i$ is an inner product on $\mathbb{R}^n$. It is bilinear and symmetric by the rules of matrix multiplication, and $\mathbf{x}^\top \mathbf{x} = x_1^2 + \cdots + x_n^2 > 0$ for $\mathbf{x} \neq \mathbf{0}$. The space $\mathbb{R}^n$ with the dot product is called the Euclidean space. Unless we say otherwise, we always use the dot product on $\mathbb{R}^n$.

Example 2.3. On $\mathbb{R}^2$, define

$$\langle \mathbf{x}, \mathbf{y}\rangle = x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2x_2 y_2.$$

This function is bilinear and symmetric. It is also positive definite, since

$$\langle \mathbf{x}, \mathbf{x}\rangle = x_1^2 - 2x_1 x_2 + 2x_2^2 = (x_1 - x_2)^2 + x_2^2 \geq 0,$$

with equality only if $x_2 = 0$ and $x_1 = x_2$, i.e., $\mathbf{x} = \mathbf{0}$. So it is an inner product, different from the dot product: for example, for $\mathbf{x} = (1, 1)^\top$, we get $\langle \mathbf{x}, \mathbf{x}\rangle = 1$, while $\mathbf{x}^\top \mathbf{x} = 2$.

Example 2.4.

- For positive weights $w_1, \ldots, w_n > 0$, the weighted dot product $\langle \mathbf{x}, \mathbf{y}\rangle = w_1 x_1 y_1 + \cdots + w_n x_n y_n$ is an inner product on $\mathbb{R}^n$. In machine learning, such weights can be used to make some features more important than others.
- The function $\langle \mathbf{x}, \mathbf{y}\rangle = x_1 y_1 - x_2 y_2$ on $\mathbb{R}^2$ is bilinear and symmetric, but it is not an inner product, since $\langle \mathbf{e}_2, \mathbf{e}_2\rangle = -1 < 0$.
- On the space $\mathbb{R}^{m \times n}$ of matrices, $\langle A, B\rangle = \operatorname{tr}(A^\top B) = \sum_{i,j} A_{ij} B_{ij}$ is an inner product, the Frobenius inner product. It is the dot product of the lists of entries of A and B.

2.1 Symmetric positive definite matrices

All the inner products on $\mathbb{R}^n$ in the examples above can be written using a matrix. For instance, the inner product of Example 2.3 is

$$\langle \mathbf{x}, \mathbf{y}\rangle = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \mathbf{x}^\top A \mathbf{y}.$$

The matrices that give inner products in this way are the following.

Definition 2.5 (Symmetric positive definite matrix)**.** A symmetric $n \times n$ matrix A is called positive definite if

$$\mathbf{x}^\top A \mathbf{x} > 0 \qquad \text{for all } \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{x} \neq \mathbf{0},$$

and positive semidefinite if $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$. A symmetric positive definite matrix is also called an SPD matrix.

Theorem 2.6. *1. If A is a symmetric positive definite $n \times n$ matrix, then $\langle \mathbf{x}, \mathbf{y}\rangle_A = \mathbf{x}^\top A \mathbf{y}$ is an inner product on $\mathbb{R}^n$.*

2. *Conversely, let $\langle \cdot, \cdot \rangle$ be an inner product on an n-dimensional vector space V, and let $\mathcal{B} = (b_1, \ldots, b_n)$ be an ordered basis of V. Let A be the $n \times n$ matrix with entries $A_{ij} = \langle b_i, b_j \rangle$. Then A is symmetric positive definite, and*

$$\langle x, y \rangle = [x]_{\mathcal{B}}^{\top} A \, [y]_{\mathcal{B}} \qquad \text{for all } x, y \in V,$$

where $[x]_{\mathcal{B}}$ is the coordinate vector of x with respect to $\mathcal{B}$, as in Lecture 1.

Proof. (1) The function $x^{\top} A y$ is bilinear by the distributivity of matrix multiplication and the rules for scalars. It is symmetric, since $y^{\top} A x$ is a 1×1 matrix and therefore equal to its transpose $(y^{\top} A x)^{\top} = x^{\top} A^{\top} y = x^{\top} A y$. It is positive definite by assumption.

(2) Write $x = \sum_i \psi_i b_i$ and $y = \sum_j \lambda_j b_j$, so that $[x]_{\mathcal{B}} = (\psi_1, \ldots, \psi_n)^{\top}$ and $[y]_{\mathcal{B}} = (\lambda_1, \ldots, \lambda_n)^{\top}$. By bilinearity,

$$\langle x, y \rangle = \Big\langle \sum_i \psi_i b_i, \sum_j \lambda_j b_j \Big\rangle = \sum_{i,j} \psi_i \langle b_i, b_j \rangle \lambda_j = [x]_{\mathcal{B}}^{\top} A [y]_{\mathcal{B}}.$$

The matrix A is symmetric, since $\langle b_i, b_j \rangle = \langle b_j, b_i \rangle$. If $\boldsymbol{\psi} \in \mathbb{R}^n$ is nonzero, then $x = \sum_i \psi_i b_i \neq 0$, since the b_i are linearly independent, and hence $\boldsymbol{\psi}^{\top} A \boldsymbol{\psi} = \langle x, x \rangle > 0$. □

The matrix A in part 2 is called the Gram matrix of the basis $\mathcal{B}$. So, once we fix a basis, inner products on an n-dimensional space are the same thing as symmetric positive definite $n \times n$ matrices. The dot product on $\mathbb{R}^n$ corresponds to $A = I_n$ and the standard basis.

Proposition 2.7. *Let A be a symmetric positive definite $n \times n$ matrix.*

1. *The diagonal entries of A are positive.*
2. *A is invertible.*
3. *For $n = 2$: a symmetric matrix $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is positive definite if and only if $a > 0$ and $\det(A) = ac - b^2 > 0$.*

Proof. (1) $A_{ii} = \mathbf{e}_i^{\top} A \mathbf{e}_i > 0$.

(2) If $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}^{\top} A \mathbf{x} = 0$, so $\mathbf{x} = \mathbf{0}$. Hence $\operatorname{null}(A) = \{\mathbf{0}\}$, and A is invertible by Lecture 1.

(3) If A is positive definite, then $a > 0$ by part 1. If $a > 0$, completing the square gives

$$\mathbf{x}^{\top} A \mathbf{x} = a x_1^2 + 2 b x_1 x_2 + c x_2^2 = a\Big(x_1 + \frac{b}{a} x_2\Big)^2 + \frac{ac - b^2}{a} x_2^2.$$

If $ac - b^2 > 0$, both terms are nonnegative, and they are both zero only if $x_2 = 0$ and $x_1 = 0$; so A is positive definite. If $ac - b^2 \leq 0$, then $\mathbf{x} = (-b, a)^{\top} \neq \mathbf{0}$ gives $\mathbf{x}^{\top} A \mathbf{x} = (ac - b^2) a \leq 0$, so A is not positive definite. □

For larger matrices, there is a similar test, Sylvester's criterion: a symmetric matrix is positive definite if and only if the determinants of all its upper left $k \times k$ submatrices, for $k = 1, \ldots, n$, are positive. We prove it in Section 10 (Corollary 10.4), together with another test, using eigenvalues (Corollary 10.3).

Example 2.8. The matrix $A_1 = \begin{pmatrix} 9 & 6 \\ 6 & 5 \end{pmatrix}$ is positive definite, since $9 > 0$ and $\det(A_1) = 45 - 36 = 9 > 0$. Indeed, $\mathbf{x}^{\top} A_1 \mathbf{x} = (3x_1 + 2x_2)^2 + x_2^2$. The matrix $A_2 = \begin{pmatrix} 9 & 6 \\ 6 & 3 \end{pmatrix}$ is not, since $\det(A_2) = 27 - 36 = -9 < 0$: for $\mathbf{x} = (2, -3)^{\top}$, we get $\mathbf{x}^{\top} A_2 \mathbf{x} = -9$.

Example 2.9. Let B be an $m \times n$ matrix with linearly independent columns. Then $B^\top B$ is symmetric positive definite: it is symmetric (Lecture 1), and for $\mathbf{x} \neq \mathbf{0}$ we have $B\mathbf{x} \neq \mathbf{0}$, so

$$\mathbf{x}^\top B^\top B\mathbf{x} = (B\mathbf{x})^\top (B\mathbf{x}) = \|B\mathbf{x}\|_2^2 > 0.$$

Matrices of the form $B^\top B$ appear everywhere in machine learning, for example in linear regression (Section 8.3) and as (scaled) covariance matrices of data.

Conversely, every symmetric positive definite matrix is of the form $B^\top B$, even with a triangular B. This is the most efficient way to compute with such matrices.

Theorem 2.10 (Cholesky decomposition)**.** *A symmetric $n \times n$ matrix A is positive definite if and only if $A = LL^\top$ for a lower triangular matrix L with positive diagonal entries. In this case, L is unique.*

Proof. If $A = LL^\top$, then $L^\top$ is invertible (it is triangular, with nonzero diagonal entries), so $\mathbf{x}^\top A\mathbf{x} = \|L^\top \mathbf{x}\|^2 > 0$ for $\mathbf{x} \neq \mathbf{0}$. Conversely, we use induction on n; for $n = 1$, $A = (a)$ with $a > 0$, and $L = (\sqrt{a})$ is the only choice. For $n \geq 2$, write

$$A = \begin{pmatrix} a & \mathbf{b}^\top \\ \mathbf{b} & A' \end{pmatrix}, \qquad \text{with } a = A_{11} > 0 \text{ (Proposition 2.7)},$$

and let $S = A' - \frac{1}{a}\mathbf{b}\mathbf{b}^\top$ be the Schur complement of a (Lecture 1). For $\mathbf{y} \in \mathbb{R}^{n-1}$ with $\mathbf{y} \neq \mathbf{0}$, the vector $\mathbf{x} = (-\frac{1}{a}\mathbf{b}^\top \mathbf{y}, \mathbf{y})$ is nonzero, and expanding gives $0 < \mathbf{x}^\top A\mathbf{x} = \frac{1}{a}(\mathbf{b}^\top \mathbf{y})^2 - \frac{2}{a}(\mathbf{b}^\top \mathbf{y})^2 + \mathbf{y}^\top A'\mathbf{y} = \mathbf{y}^\top S\mathbf{y}$. So S is symmetric positive definite, and by induction, $S = L'L'^\top$ with a unique lower triangular L' with positive diagonal entries. Then

$$L = \begin{pmatrix} \sqrt{a} & \mathbf{0}^\top \\ \frac{1}{\sqrt{a}}\mathbf{b} & L' \end{pmatrix} \qquad \text{satisfies} \qquad LL^\top = \begin{pmatrix} a & \mathbf{b}^\top \\ \mathbf{b} & \frac{1}{a}\mathbf{b}\mathbf{b}^\top + L'L'^\top \end{pmatrix} = A.$$

For uniqueness, compare the blocks of $LL^\top = A$ for any such L: the upper left entry gives $L_{11}^2 = a$, so $L_{11} = \sqrt{a}$; the first column gives L_{11} times the rest of the first column of L equal to $\mathbf{b}$; and the lower right block then gives $L'L'^\top = S$ for the lower right block L' of L, which determines L' by induction. □

The proof is an algorithm: take the square root of the upper left entry, divide the rest of the first column by it, and continue with the Schur complement. It is the LU decomposition of Lecture 1 with $U = L^\top$, and it needs about half as many operations. It fails, with a nonpositive number under a square root, exactly when A is not positive definite, which is how `np.linalg.cholesky` detects such matrices. Once $A = LL^\top$ is known, $A\mathbf{x} = \mathbf{b}$ is solved by forward substitution in $L\mathbf{y} = \mathbf{b}$ and back substitution in $L^\top \mathbf{x} = \mathbf{y}$, and $\det(A) = (L_{11} \cdots L_{nn})^2$. Lecture 4 uses the Cholesky decomposition to sample Gaussian vectors and Gaussian processes.

Example 2.11. For $A = \begin{pmatrix} 9 & 6 \\ 6 & 5 \end{pmatrix}$, the algorithm gives $L_{11} = 3$, $L_{21} = \frac{6}{3} = 2$ and $L_{22} = \sqrt{5 - 2^2} = 1$, so

$$A = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{x}^\top A\mathbf{x} = \|L^\top \mathbf{x}\|^2 = (3x_1 + 2x_2)^2 + x_2^2.$$

The Cholesky decomposition is a systematic way of completing the square, as in the proof of Proposition 2.7.

Remark 2.12 (Kernels). Many machine learning methods, such as support vector machines, only use the data through inner products $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ of data points. Kernel methods replace them by $k(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$, the inner product of the data after a (possibly nonlinear) feature map ϕ. The matrix K with entries $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ is then a Gram matrix, and it is symmetric positive semidefinite.

3 Lengths and Distances

3.1 The norm of an inner product

Every inner product gives a norm, in the same way that the dot product gives the Euclidean norm $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}}$.

Definition 3.1 (Induced norm)**.** Let V be an inner product space. The length, or norm, of a vector $x \in V$ is

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

A vector x with $\|x\| = 1$ is called a unit vector. For $x \neq 0$, the vector $\frac{x}{\|x\|}$ is a unit vector; replacing x by $\frac{x}{\|x\|}$ is called normalizing x.

Example 3.2. For $\mathbf{x} = (1, 1)^\top$, the dot product gives $\|\mathbf{x}\| = \sqrt{2}$, while the inner product of Example 2.3 gives $\|\mathbf{x}\| = \sqrt{1} = 1$. So the length of a vector depends on the choice of the inner product.

To show that $\|x\| = \sqrt{\langle x, x \rangle}$ really is a norm, we need the most important inequality of linear algebra.

Theorem 3.3 (Cauchy–Schwarz inequality)**.** *Let V be an inner product space. For all $x, y \in V$,*

$$|\langle x, y \rangle| \leq \|x\| \, \|y\|,$$

with equality if and only if x and y are linearly dependent.

Proof. If $y = 0$, both sides are 0, and x, y are linearly dependent. So let $y \neq 0$. For every $t \in \mathbb{R}$, by bilinearity and symmetry,

$$0 \leq \langle x - ty, x - ty \rangle = \|x\|^2 - 2t\langle x, y \rangle + t^2 \|y\|^2.$$

Choosing $t = \frac{\langle x, y \rangle}{\|y\|^2}$ gives

$$0 \leq \|x\|^2 - 2\frac{\langle x, y \rangle^2}{\|y\|^2} + \frac{\langle x, y \rangle^2}{\|y\|^2} = \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2},$$

i.e., $\langle x, y \rangle^2 \leq \|x\|^2 \|y\|^2$. Taking square roots proves the inequality. Equality holds if and only if $\langle x - ty, x - ty \rangle = 0$ for this t, i.e., $x = ty$, which (for $y \neq 0$) means that x and y are linearly dependent. □

Corollary 3.4. *If V is an inner product space, then $\|x\| = \sqrt{\langle x, x \rangle}$ is a norm on V. In particular, the Euclidean norm $\|\cdot\|_2$ is a norm on $\mathbb{R}^n$.*

Proof. Positive definiteness follows from the positive definiteness of the inner product, and $\|\lambda x\| = \sqrt{\lambda^2 \langle x, x \rangle} = |\lambda| \|x\|$. For the triangle inequality, by the Cauchy–Schwarz inequality,

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2. \qquad \square$$

Example 3.5. Let $\mathbf{a} = (|x_1|, \dots, |x_n|)^\top$ and $\mathbf{1} = (1, \dots, 1)^\top$. Applying the Cauchy–Schwarz inequality for the dot product to $\mathbf{a}$ and $\mathbf{1}$ gives

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i| \cdot 1 \le \sqrt{\sum_{i=1}^{n} x_i^2} \sqrt{\sum_{i=1}^{n} 1^2} = \sqrt{n}\, \|\mathbf{x}\|_2.$$

This improves the inequality $\|\mathbf{x}\|_1 \le n\|\mathbf{x}\|_\infty$ of Proposition 1.5, since $\sqrt{n}\|\mathbf{x}\|_2 \le n\|\mathbf{x}\|_\infty$.

Not every norm comes from an inner product. Norms that do come from an inner product can be recognized by the following identities.

Proposition 3.6 (Parallelogram law and polarization)**.** *Let V be an inner product space with the induced norm. For all $x, y \in V$:*

1. $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ *(parallelogram law);*
2. $\langle x, y\rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2)$ *(polarization identity).*

Proof. Expanding with bilinearity, $\|x \pm y\|^2 = \|x\|^2 \pm 2\langle x, y\rangle + \|y\|^2$. Adding the two identities gives part 1; subtracting them gives part 2. □

The parallelogram law says that in a parallelogram, the sum of the squares of the two diagonals equals the sum of the squares of the four sides. The polarization identity shows that the inner product can be recovered from its norm.

Example 3.7. The ℓ_1 norm on $\mathbb{R}^2$ does not come from any inner product: for $\mathbf{x} = \mathbf{e}_1$ and $\mathbf{y} = \mathbf{e}_2$, we get $\|\mathbf{x} + \mathbf{y}\|_1^2 + \|\mathbf{x} - \mathbf{y}\|_1^2 = 4 + 4 = 8$, but $2\|\mathbf{x}\|_1^2 + 2\|\mathbf{y}\|_1^2 = 4$, so the parallelogram law fails. Hence normed spaces are more general than inner product spaces.

3.2 Distances

Definition 3.8 (Distance)**.** Let V be an inner product space (or, more generally, a normed space). The distance between $x, y \in V$ is

$$d(x, y) = \|x - y\|.$$

For the dot product on $\mathbb{R}^n$, this is the Euclidean distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_i (x_i - y_i)^2}$.

Proposition 3.9. *The distance d has the following properties, for all $x, y, z \in V$:*

1. *Positive definiteness: $d(x, y) \ge 0$, and $d(x, y) = 0$ if and only if $x = y$;*
2. *Symmetry: $d(x, y) = d(y, x)$;*
3. *Triangle inequality: $d(x, z) \le d(x, y) + d(y, z)$.*

Proof. Part 1 follows from the positive definiteness of the norm, and part 2 from $\|x - y\| = \|(-1)(y - x)\| = \|y - x\|$. Part 3 is the triangle inequality for the norm, applied to $x - z = (x - y) + (y - z)$. □

A function $d : X \times X \to \mathbb{R}$ on any set X with these three properties is called a metric, and (X, d) is called a metric space. So every normed space is a metric space.

Example 3.10. For $\mathbf{x} = (1,2)^\top$ and $\mathbf{y} = (4,6)^\top$, we have $\mathbf{x} - \mathbf{y} = (-3,-4)^\top$, so the distance between $\mathbf{x}$ and $\mathbf{y}$ is

$$\|\mathbf{x}-\mathbf{y}\|_2 = \sqrt{9+16} = 5, \qquad \|\mathbf{x}-\mathbf{y}\|_1 = 3+4 = 7, \qquad \|\mathbf{x}-\mathbf{y}\|_\infty = 4$$

for the Euclidean, Manhattan and maximum norms, respectively.

Remark 3.11 (Distances in machine learning). The k-nearest neighbor method classifies a new data point by a majority vote among the k training points that are closest to it. Which points are closest depends on the metric, so the choice of the norm matters. It also depends on the units of the features: if one feature is measured in meters and another in millimeters, the second one dominates the Euclidean distance. This is why features are usually standardized first, or, equivalently, a weighted inner product is used.

4 Angles and Orthogonality

4.1 Angles

By the Cauchy–Schwarz inequality, $-1 \le \frac{\langle x,y\rangle}{\|x\|\|y\|} \le 1$ for nonzero vectors x and y. So the following definition makes sense.

Definition 4.1 (Angle)**.** Let V be an inner product space, and let $x, y \in V$ be nonzero. The angle between x and y is the unique number $\omega \in [0,\pi]$ with

$$\cos\omega = \frac{\langle x,y\rangle}{\|x\|\,\|y\|}.$$

For the dot product in $\mathbb{R}^2$ or $\mathbb{R}^3$, this is the usual angle between two arrows. Indeed, expanding $\|\mathbf{x}-\mathbf{y}\|^2$ gives

$$\|\mathbf{x}-\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\langle\mathbf{x},\mathbf{y}\rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\|\|\mathbf{y}\|\cos\omega,$$

which is the law of cosines for the triangle with sides $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{x}-\mathbf{y}$ (see Figure 2).

Example 4.2. The angle ω between $\mathbf{x} = (3,1)^\top$ and $\mathbf{y} = (1,2)^\top$ satisfies

$$\cos\omega = \frac{3\cdot 1 + 1\cdot 2}{\sqrt{10}\,\sqrt{5}} = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}},$$

so $\omega = \frac{\pi}{4}$, i.e., $45°$.

4.2 Orthogonality

Definition 4.3 (Orthogonality)**.** Two vectors x and y of an inner product space are orthogonal, written $x \perp y$, if $\langle x,y\rangle = 0$. If, in addition, $\|x\| = \|y\| = 1$, then x and y are called orthonormal.

So two nonzero vectors are orthogonal if and only if the angle between them is $\frac{\pi}{2}$. The zero vector is orthogonal to every vector.

Example 4.4. The vectors $\mathbf{x} = (1,1)^\top$ and $\mathbf{y} = (-1,1)^\top$ are orthogonal for the dot product, since $\mathbf{x}^\top\mathbf{y} = -1+1 = 0$. Now consider the inner product $\langle\mathbf{x},\mathbf{y}\rangle = \mathbf{x}^\top A\mathbf{y}$ with $A = \operatorname{diag}(2,1)$. Then

$$\langle\mathbf{x},\mathbf{y}\rangle = 2\cdot 1\cdot(-1) + 1\cdot 1 = -1, \qquad \|\mathbf{x}\|^2 = 2+1 = 3, \qquad \|\mathbf{y}\|^2 = 2+1 = 3,$$

so $\cos\omega = -\frac{1}{3}$ and $\omega \approx 1.91 \approx 109.5°$. So whether two vectors are orthogonal depends on the inner product.

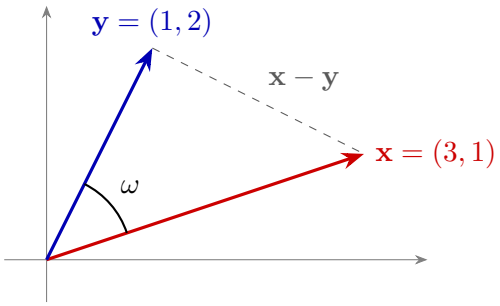


Figure 2: The angle $\omega = 45°$ between $\mathbf{x} = (3, 1)$ and $\mathbf{y} = (1, 2)$ (Example 4.2). The triangle with sides $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{x} - \mathbf{y}$ satisfies the law of cosines.

Theorem 4.5 (Pythagoras)**.** *If $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.*

Proof. $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 = \|x\|^2 + \|y\|^2$. □

Remark 4.6 (Cosine similarity). In machine learning, the number $\cos\omega = \frac{\langle \mathbf{x},\mathbf{y}\rangle}{\|\mathbf{x}\|\|\mathbf{y}\|}$ is called the cosine similarity of $\mathbf{x}$ and $\mathbf{y}$. It is 1 if $\mathbf{x}$ and $\mathbf{y}$ point in the same direction, 0 if they are orthogonal, and -1 if they point in opposite directions. Unlike the distance, it does not change when $\mathbf{x}$ or $\mathbf{y}$ is multiplied by a positive scalar. This makes it useful for comparing text documents (represented by word counts, where long and short documents should be comparable) and embeddings, e.g., the vectors that language models assign to words and sentences.

4.3 Orthogonal matrices

In Lecture 1, we defined an $n \times n$ matrix Q to be orthogonal if $Q^\top Q = I_n$, and we claimed that orthogonal matrices preserve lengths and angles. We can now make this precise and prove it.

Theorem 4.7. *For an $n \times n$ matrix Q, the following statements are equivalent:*

1. *Q is orthogonal, i.e., $Q^\top Q = I_n$;*
2. *Q preserves the dot product: $(Q\mathbf{x})^\top (Q\mathbf{y}) = \mathbf{x}^\top \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$;*
3. *Q preserves lengths: $\|Q\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x} \in \mathbb{R}^n$;*
4. *the columns of Q are orthonormal.*

In particular, orthogonal matrices preserve lengths, distances and angles.

Proof. $(1 \Rightarrow 2)$ $(Q\mathbf{x})^\top (Q\mathbf{y}) = \mathbf{x}^\top Q^\top Q\mathbf{y} = \mathbf{x}^\top \mathbf{y}$.

$(2 \Rightarrow 3)$ Take $\mathbf{y} = \mathbf{x}$ and take square roots.

$(3 \Rightarrow 2)$ By the polarization identity (Proposition 3.6) and linearity,

$$(Q\mathbf{x})^\top (Q\mathbf{y}) = \tfrac{1}{4}(\|Q\mathbf{x} + Q\mathbf{y}\|^2 - \|Q\mathbf{x} - Q\mathbf{y}\|^2) = \tfrac{1}{4}(\|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2) = \mathbf{x}^\top \mathbf{y}.$$

$(2 \Rightarrow 1)$ The (i, j) entry of $Q^\top Q$ is $\mathbf{e}_i^\top Q^\top Q\mathbf{e}_j = (Q\mathbf{e}_i)^\top (Q\mathbf{e}_j) = \mathbf{e}_i^\top \mathbf{e}_j$, which is 1 if $i = j$ and 0 otherwise.

$(1 \Leftrightarrow 4)$ The (i, j) entry of $Q^\top Q$ is $\mathbf{q}_i^\top \mathbf{q}_j$, where $\mathbf{q}_1, \ldots, \mathbf{q}_n$ are the columns of Q. So $Q^\top Q = I_n$ means exactly that $\mathbf{q}_i^\top \mathbf{q}_j$ is 1 for $i = j$ and 0 for $i \neq j$.

Finally, distances are lengths of differences, $\|Q\mathbf{x} - Q\mathbf{y}\|_2 = \|Q(\mathbf{x} - \mathbf{y})\|_2 = \|\mathbf{x} - \mathbf{y}\|_2$, and angles are defined in terms of the dot product and lengths. □

A map that preserves distances is called an isometry. Rotations and reflections are the typical examples; we study rotations in Section 9.

5 Orthonormal Bases

In $\mathbb{R}^n$, the standard basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ consists of vectors of length 1 that are pairwise orthogonal. Bases with this property are especially convenient.

Definition 5.1 (Orthonormal basis)**.** Let V be an inner product space. Vectors $b_1, \dots, b_k \in V$ are called orthogonal if $\langle b_i, b_j \rangle = 0$ for all $i \neq j$, and orthonormal if, in addition, $\|b_i\| = 1$ for all i. In other words, $b_1, \dots, b_k$ are orthonormal if

$$\langle b_i, b_j \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

An orthonormal basis (ONB) of V is a basis of V consisting of orthonormal vectors.

Lemma 5.2. *Orthogonal nonzero vectors $b_1, \dots, b_k$ are linearly independent.*

Proof. Suppose that $a_1 b_1 + \cdots + a_k b_k = 0$. Taking the inner product with b_j and using bilinearity gives

$$0 = \langle a_1 b_1 + \cdots + a_k b_k, b_j \rangle = a_1 \langle b_1, b_j \rangle + \cdots + a_k \langle b_k, b_j \rangle = a_j \|b_j\|^2.$$

Since $b_j \neq 0$, we get $a_j = 0$, for every j. □

The main advantage of an orthonormal basis is that coordinates are computed by inner products, without solving a system of linear equations.

Proposition 5.3. *Let $b_1, \dots, b_n$ be an orthonormal basis of V. Then for all $u, v \in V$:*

1. $v = \langle v, b_1 \rangle b_1 + \langle v, b_2 \rangle b_2 + \cdots + \langle v, b_n \rangle b_n$;
2. $\langle u, v \rangle = \sum_{i=1}^n \langle u, b_i \rangle \langle v, b_i \rangle$, *and in particular* $\|v\|^2 = \sum_{i=1}^n \langle v, b_i \rangle^2$ *(Parseval's identity).*

Proof. (1) Write $v = c_1 b_1 + \cdots + c_n b_n$. Taking the inner product with b_j gives $\langle v, b_j \rangle = \sum_i c_i \langle b_i, b_j \rangle = c_j$.

(2) Write $u = \sum_i d_i b_i$ and $v = \sum_j c_j b_j$ with $d_i = \langle u, b_i \rangle$ and $c_j = \langle v, b_j \rangle$. Then $\langle u, v \rangle = \sum_{i,j} d_i c_j \langle b_i, b_j \rangle = \sum_i d_i c_i$. □

In the language of Theorem 2.6: the Gram matrix of an orthonormal basis is the identity matrix, so in coordinates with respect to an orthonormal basis, every inner product becomes the dot product.

Example 5.4. The vectors $\mathbf{b}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$ and $\mathbf{b}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$ form an orthonormal basis of $\mathbb{R}^2$. For $\mathbf{v} = (3,1)^\top$, we get

$$\langle \mathbf{v}, \mathbf{b}_1 \rangle = \frac{4}{\sqrt{2}} = 2\sqrt{2}, \qquad \langle \mathbf{v}, \mathbf{b}_2 \rangle = \frac{2}{\sqrt{2}} = \sqrt{2},$$

so $\mathbf{v} = 2\sqrt{2}\,\mathbf{b}_1 + \sqrt{2}\,\mathbf{b}_2 = 2(1,1)^\top + (1,-1)^\top$. Parseval's identity checks out: $(2\sqrt{2})^2 + (\sqrt{2})^2 = 10 = 3^2 + 1^2$.

Remark 5.5. If $\mathbf{b}_1, \dots, \mathbf{b}_n$ is an orthonormal basis of $\mathbb{R}^n$ (with the dot product), then the matrix Q with columns $\mathbf{b}_1, \dots, \mathbf{b}_n$ is orthogonal (Theorem 4.7). The coordinate vector of $\mathbf{v}$ is then $Q^{-1}\mathbf{v} = Q^\top \mathbf{v} = (\mathbf{b}_1^\top \mathbf{v}, \dots, \mathbf{b}_n^\top \mathbf{v})^\top$, in agreement with Proposition 5.3.

5.1 The Gram–Schmidt process

Every basis can be turned into an orthonormal basis. The idea is to go through the basis vectors one by one, and to subtract from each vector its components in the directions of the previous ones.

Theorem 5.6 (Gram–Schmidt process)**.** *Let $v_1, \dots, v_k$ be linearly independent vectors in an inner product space V. Define recursively, for $j = 1, \dots, k$,*

$$u_j = v_j - \sum_{i=1}^{j-1} \langle v_j, e_i \rangle \, e_i, \qquad e_j = \frac{u_j}{\|u_j\|}.$$

Then $e_1, \dots, e_k$ are orthonormal, and $\operatorname{span}\{e_1, \dots, e_j\} = \operatorname{span}\{v_1, \dots, v_j\}$ *for every $j = 1, \dots, k$.*

Proof. We use induction on j. For $j = 1$, we have $u_1 = v_1 \neq 0$ and $e_1 = v_1/\|v_1\|$, so the claim holds. Now let $j \geq 2$, and assume that $e_1, \dots, e_{j-1}$ are orthonormal with $\operatorname{span}\{e_1, \dots, e_{j-1}\} = \operatorname{span}\{v_1, \dots, v_{j-1}\}$.

First, $u_j \neq 0$: otherwise $v_j = \sum_{i<j} \langle v_j, e_i \rangle e_i$ would lie in $\operatorname{span}\{e_1, \dots, e_{j-1}\}$, which equals $\operatorname{span}\{v_1, \dots, v_{j-1}\}$, contradicting the linear independence of $v_1, \dots, v_j$. So e_j is well defined and has length 1.

Next, for $l < j$, using that $e_1, \dots, e_{j-1}$ are orthonormal,

$$\langle u_j, e_l \rangle = \langle v_j, e_l \rangle - \sum_{i<j} \langle v_j, e_i \rangle \langle e_i, e_l \rangle = \langle v_j, e_l \rangle - \langle v_j, e_l \rangle = 0,$$

so e_j is orthogonal to $e_1, \dots, e_{j-1}$. Finally, e_j is a linear combination of v_j and $e_1, \dots, e_{j-1}$, and conversely $v_j = \|u_j\| e_j + \sum_{i<j} \langle v_j, e_i \rangle e_i$. Together with the induction hypothesis, this gives $\operatorname{span}\{e_1, \dots, e_j\} = \operatorname{span}\{v_1, \dots, v_j\}$. □

The vector $\langle v_j, e_i \rangle e_i$ is the component of v_j in the direction of e_i; in Section 8, we will call it the orthogonal projection of v_j onto $\operatorname{span}\{e_i\}$. In hand computations, it is often easier to postpone the normalization: since $\langle v_j, e_i \rangle e_i = \frac{\langle v_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i$, the vectors

$$u_j = v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, u_i \rangle}{\langle u_i, u_i \rangle} \, u_i$$

are orthogonal, and we normalize them at the end. Figure 3 illustrates the process for two vectors.

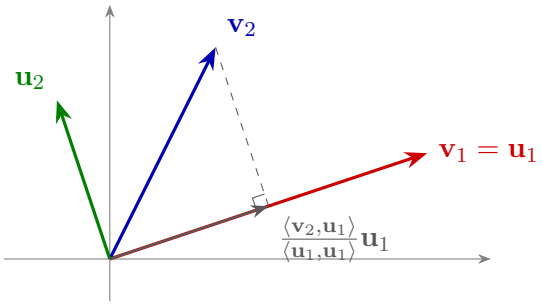


Figure 3: The Gram–Schmidt process for $\mathbf{v}_1 = (3, 1)$ and $\mathbf{v}_2 = (1, 2)$. The component of $\mathbf{v}_2$ in the direction of $\mathbf{u}_1 = \mathbf{v}_1$ is $\frac{5}{10}\mathbf{u}_1 = (1.5, 0.5)$; subtracting it gives $\mathbf{u}_2 = (-0.5, 1.5)$, which is orthogonal to $\mathbf{u}_1$. Normalizing $\mathbf{u}_1$ and $\mathbf{u}_2$ gives an orthonormal basis of $\mathbb{R}^2$.

Example 5.7. Let us apply the Gram–Schmidt process to the basis $\mathbf{v}_1 = (1,1,0)^\top$, $\mathbf{v}_2 = (1,0,1)^\top$, $\mathbf{v}_3 = (0,1,1)^\top$ of $\mathbb{R}^3$, using the unnormalized version:

$$\mathbf{u}_1 = \mathbf{v}_1 = (1,1,0)^\top,$$
$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2^\top \mathbf{u}_1}{\mathbf{u}_1^\top \mathbf{u}_1}\mathbf{u}_1 = (1,0,1)^\top - \frac{1}{2}(1,1,0)^\top = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)^\top,$$
$$\mathbf{u}_3 = \mathbf{v}_3 - \frac{\mathbf{v}_3^\top \mathbf{u}_1}{\mathbf{u}_1^\top \mathbf{u}_1}\mathbf{u}_1 - \frac{\mathbf{v}_3^\top \mathbf{u}_2}{\mathbf{u}_2^\top \mathbf{u}_2}\mathbf{u}_2 = (0,1,1)^\top - \frac{1}{2}(1,1,0)^\top - \frac{1/2}{3/2}\left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)^\top = \left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right)^\top.$$

One checks that $\mathbf{u}_1^\top \mathbf{u}_2 = \mathbf{u}_1^\top \mathbf{u}_3 = \mathbf{u}_2^\top \mathbf{u}_3 = 0$. Normalizing gives the orthonormal basis

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}, \qquad \mathbf{e}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\-1\\2\end{pmatrix}, \qquad \mathbf{e}_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}.$$

Corollary 5.8. *Every finite-dimensional inner product space has an orthonormal basis.*

Proof. Apply the Gram–Schmidt process to any basis $v_1, \dots, v_n$. The resulting vectors $e_1, \dots, e_n$ are orthonormal and span $\operatorname{span}\{v_1, \dots, v_n\} = V$, and they are linearly independent by Lemma 5.2. □

The Gram–Schmidt process can also be written as a factorization of matrices.

Proposition 5.9 (QR decomposition)**.** *Let A be an $m \times n$ matrix with linearly independent columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Then $A = QR$, where Q is an $m \times n$ matrix with orthonormal columns (i.e., $Q^\top Q = I_n$), and R is an upper triangular $n \times n$ matrix with positive diagonal entries.*

Proof. Apply the Gram–Schmidt process to $\mathbf{a}_1, \dots, \mathbf{a}_n$ (with the dot product), and let Q be the matrix with columns $\mathbf{e}_1, \dots, \mathbf{e}_n$. Since $\mathbf{a}_j \in \operatorname{span}\{\mathbf{e}_1, \dots, \mathbf{e}_j\}$ and $\mathbf{e}_1, \dots, \mathbf{e}_j$ are orthonormal, the argument of Proposition 5.3 gives $\mathbf{a}_j = \sum_{i=1}^{j} (\mathbf{e}_i^\top \mathbf{a}_j)\, \mathbf{e}_i$. So $A = QR$, where $R_{ij} = \mathbf{e}_i^\top \mathbf{a}_j$ for $i \le j$ and $R_{ij} = 0$ for $i > j$. The diagonal entries are $R_{jj} = \mathbf{e}_j^\top \mathbf{a}_j = \mathbf{e}_j^\top \mathbf{u}_j = \|\mathbf{u}_j\| > 0$, since $\mathbf{a}_j - \mathbf{u}_j$ is a linear combination of $\mathbf{e}_1, \dots, \mathbf{e}_{j-1}$, which are orthogonal to $\mathbf{e}_j$. □

Example 5.10. For the matrix A with columns v_1, v_2, v_3 of Example 5.7, the matrix Q has columns $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and

$$R = Q^\top A = \begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{\sqrt{6}}{2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{pmatrix}.$$

Remark 5.11. In NumPy, `Q, R = np.linalg.qr(A)` computes a QR decomposition (the signs of some columns of Q and rows of R may differ from the ones above). It does not use the Gram–Schmidt process directly, since in floating-point arithmetic the vectors $\mathbf{e}_j$ computed by the Gram–Schmidt process can be far from orthogonal; see Programming Exercise P3. The QR decomposition is used, for example, to solve least squares problems in a numerically stable way.

6 Orthogonal Complements

Definition 6.1 (Orthogonal complement)**.** Let U be a subspace of an inner product space V. The orthogonal complement of U is the set

$$U^\perp = \{v \in V : \langle v, u \rangle = 0 \text{ for all } u \in U\}$$

of all vectors that are orthogonal to every vector of U.

Lemma 6.2. *Let U be a subspace of an inner product space V.*

1. *$U^\perp$ is a subspace of V, and $U \cap U^\perp = \{0\}$.*
2. *If $U = \operatorname{span}\{u_1, \dots, u_k\}$, then $v \in U^\perp$ if and only if $\langle v, u_i\rangle = 0$ for $i = 1, \dots, k$.*

Proof. (1) $\langle 0, u\rangle = 0$ for all u, so $0 \in U^\perp$. If $v, w \in U^\perp$ and $\lambda \in \mathbb{R}$, then $\langle v+w, u\rangle = \langle v, u\rangle + \langle w, u\rangle = 0$ and $\langle \lambda v, u\rangle = \lambda\langle v, u\rangle = 0$ for all $u \in U$. If $v \in U \cap U^\perp$, then $\langle v, v\rangle = 0$, so $v = 0$.

(2) If $\langle v, u_i\rangle = 0$ for all i, then $\langle v, \sum_i a_i u_i\rangle = \sum_i a_i \langle v, u_i\rangle = 0$ for every element $\sum_i a_i u_i$ of U. The converse is clear. □

To state the main theorem, we need the following notion.

Definition 6.3 (Direct sum of subspaces)**.** Let U and W be subspaces of a vector space V. We say that V is the direct sum of U and W, and write $V = U \oplus W$, if every $v \in V$ can be written in exactly one way as $v = u + w$ with $u \in U$ and $w \in W$.

Lemma 6.4. *If $V = U \oplus W$, where U has a basis $u_1, \dots, u_k$ and W has a basis $w_1, \dots, w_l$, then $u_1, \dots, u_k, w_1, \dots, w_l$ is a basis of V. In particular,* $\dim V = \dim U + \dim W$.

Proof. These vectors span V, since every $v = u + w$ is a linear combination of them. If $\sum_i a_i u_i + \sum_j b_j w_j = 0$, then $0 = \sum_i a_i u_i + \sum_j b_j w_j$ and $0 = 0 + 0$ are two ways to write 0 as an element of U plus an element of W. By uniqueness, $\sum_i a_i u_i = 0$ and $\sum_j b_j w_j = 0$, so all a_i and b_j are 0. □

In the following theorem, we use the fact that every subspace of a finite-dimensional vector space is itself finite-dimensional, which was proved in Lecture 1 with the exchange lemma.

Theorem 6.5 (Orthogonal decomposition)**.** *Let U be a subspace of a finite-dimensional inner product space V. Then:*

1. *$V = U \oplus U^\perp$;*
2. *$\dim U^\perp = \dim V - \dim U$;*
3. *$(U^\perp)^\perp = U$.*

Proof. (1) By Corollary 5.8, U has an orthonormal basis $e_1, \dots, e_k$. For $v \in V$, let

$$u = \sum_{i=1}^{k} \langle v, e_i\rangle e_i \in U \qquad \text{and} \qquad w = v - u.$$

Then $\langle w, e_j\rangle = \langle v, e_j\rangle - \langle v, e_j\rangle = 0$ for all j, so $w \in U^\perp$ by Lemma 6.2, and $v = u + w$. This decomposition is unique: if also $v = u' + w'$ with $u' \in U$ and $w' \in U^\perp$, then $u - u' = w' - w$ lies in $U \cap U^\perp = \{0\}$, so $u = u'$ and $w = w'$.

(2) This follows from part 1 and Lemma 6.4.

(3) Every $u \in U$ is orthogonal to every vector of $U^\perp$, so $U \subseteq (U^\perp)^\perp$. Conversely, let $v \in (U^\perp)^\perp$, and write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Then $0 = \langle v, w\rangle = \langle u, w\rangle + \langle w, w\rangle = \|w\|^2$, so $w = 0$ and $v = u \in U$. □

Example 6.6. Let $U = \operatorname{span}\{(1,1,1)^\top\}$ in $\mathbb{R}^3$. By Lemma 6.2, $U^\perp = \{\mathbf{x} : x_1 + x_2 + x_3 = 0\}$ is the plane that we met as an example of a subspace in Lecture 1, and indeed $\dim U + \dim U^\perp = 1 + 2 = 3$. For $\mathbf{v} = (1,2,3)^\top$, the proof of Theorem 6.5 with $\mathbf{e}_1 = \frac{1}{\sqrt{3}}(1,1,1)^\top$ gives the decomposition

$$\mathbf{v} = \underbrace{\frac{6}{\sqrt{3}}\mathbf{e}_1}_{(2,2,2)^\top \in U} + \underbrace{(-1,0,1)^\top}_{\in U^\perp}.$$

6.1 The four fundamental subspaces

Orthogonal complements explain how the null space, the row space and the column space of a matrix from Lecture 1 fit together.

Theorem 6.7. *Let A be an $m \times n$ matrix. Then, with respect to the dot product,*

$$\text{null}(A) = \text{row}(A)^\perp \quad \textit{in } \mathbb{R}^n \qquad \textit{and} \qquad \text{null}(A^\top) = \text{col}(A)^\perp \quad \textit{in } \mathbb{R}^m.$$

Proof. Let $\mathbf{r}_1, \dots, \mathbf{r}_m$ be the rows of A, written as column vectors. The i-th entry of $A\mathbf{x}$ is $\mathbf{r}_i^\top \mathbf{x}$, so $A\mathbf{x} = \mathbf{0}$ if and only if $\mathbf{x}$ is orthogonal to $\mathbf{r}_1, \dots, \mathbf{r}_m$, which by Lemma 6.2 means $\mathbf{x} \in \text{row}(A)^\perp$. The second statement follows by applying the first one to $A^\top$, since $\text{row}(A^\top) = \text{col}(A)$. □

Combined with Theorem 6.5, this gives $\mathbb{R}^n = \text{row}(A) \oplus \text{null}(A)$ and $\dim \text{row}(A) + \dim \text{null}(A) = n$. Since $\dim \text{row}(A) = \text{rank}(A)$ (Lecture 1), this is another proof of the rank–nullity theorem.

Example 6.8. In Lecture 1, we found that the null space of $A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}$ is spanned by $(-1, 1, 1)^\top$. Indeed, this vector is orthogonal to both rows: $-1 + 2 - 1 = 0$ and $-3 - 1 + 4 = 0$.

Remark 6.9. By Theorem 6.5(3), the second statement can also be written as $\text{col}(A) = \text{null}(A^\top)^\perp$. So the system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is orthogonal to every solution $\mathbf{y}$ of $A^\top \mathbf{y} = \mathbf{0}$.

6.2 Hyperplanes and linear classifiers

Definition 6.10 (Hyperplane)**.** Let $\mathbf{w} \in \mathbb{R}^n$ be nonzero and $c \in \mathbb{R}$. The set

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{w}^\top \mathbf{x} = c\}$$

is called a hyperplane, and $\mathbf{w}$ is called a normal vector of H. If $c = 0$, then $H = \text{span}\{\mathbf{w}\}^\perp$ is a subspace of dimension $n - 1$ (Theorem 6.5).

If $c \neq 0$, the hyperplane H does not contain $\mathbf{0}$, so it is not a subspace. But if $\mathbf{x}_0$ is any point of H, then $\mathbf{w}^\top \mathbf{x} = c$ if and only if $\mathbf{w}^\top (\mathbf{x} - \mathbf{x}_0) = 0$. Hence

$$H = \mathbf{x}_0 + \text{span}\{\mathbf{w}\}^\perp = \{\mathbf{x}_0 + \mathbf{u} : \mathbf{u} \in \text{span}\{\mathbf{w}\}^\perp\}$$

is a shifted copy of the subspace $\text{span}\{\mathbf{w}\}^\perp$. In general, a set of the form $\mathbf{x}_0 + U$, where U is a subspace, is called an affine subspace. In $\mathbb{R}^2$, hyperplanes are lines, and in $\mathbb{R}^3$, they are planes.

Example 6.11. The plane $x + 2y + 2z = 2$ in $\mathbb{R}^3$ has the normal vector $\mathbf{w} = (1, 2, 2)^\top$ and contains the point $\mathbf{x}_0 = (2, 0, 0)^\top$. It is the shifted copy $\mathbf{x}_0 + \text{span}\{\mathbf{w}\}^\perp$ of the plane $x + 2y + 2z = 0$ through the origin.

Remark 6.12 (Linear classifiers). A linear classifier assigns to a data point $\mathbf{x} \in \mathbb{R}^n$ the class

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b) \in \{+1, -1\},$$

for a weight vector $\mathbf{w}$ and a bias $b \in \mathbb{R}$. The decision boundary $\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} + b = 0\}$ is a hyperplane with normal vector $\mathbf{w}$, and the classifier predicts $+1$ on the side into which $\mathbf{w}$ points (see Figure 4). The perceptron, logistic regression and support vector machines all learn classifiers of this form. In Section 8.4, we compute the distance of a point to the decision boundary, which measures how confident the classification is.

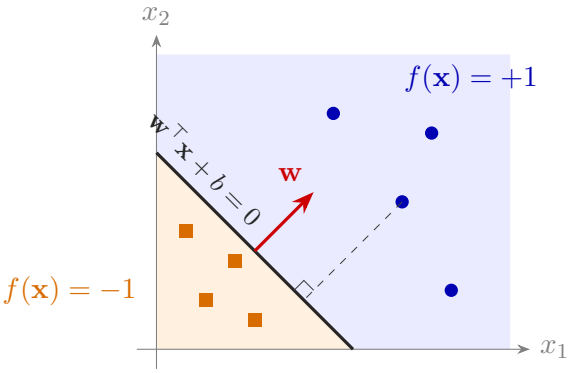


Figure 4: A linear classifier in $\mathbb{R}^2$ with $\mathbf{w} = (1, 1)$ and $b = -2$. The decision boundary is the line $x_1 + x_2 = 2$ with normal vector $\mathbf{w}$. Points on the side into which $\mathbf{w}$ points (blue disks) are classified as $+1$, the others (orange squares) as -1. The dashed segment shows the distance of the point $(2.5, 1.5)$ to the boundary, which is $\sqrt{2}$ (Example 8.15).

7 Inner Products of Functions

Inner products are not limited to $\mathbb{R}^n$. Functions can be added and multiplied by scalars (Lecture 1), and an integral gives a natural inner product on them.

Proposition 7.1. *On the vector space $C[a, b]$ of continuous functions $f : [a, b] \to \mathbb{R}$ (Lecture 1),*

$$\langle f, g \rangle = \int_a^b f(x) g(x)\, dx$$

is an inner product.

Proof. Bilinearity and symmetry follow from the linearity of the integral. For positive definiteness, $\langle f, f \rangle = \int_a^b f(x)^2\, dx \geq 0$. If $f \neq 0$, then $f(x_0) \neq 0$ for some x_0, and by continuity, $f(x)^2 \geq \frac{1}{2} f(x_0)^2$ on some interval of positive length around x_0; hence $\langle f, f \rangle > 0$. □

The continuity assumption matters: a function that is 1 at a single point and 0 everywhere else is nonzero, but the integral of its square is 0.

Example 7.2. On $C[-\pi, \pi]$, the functions sin and cos are orthogonal, since $\sin(x)\cos(x)$ is an odd function and therefore

$$\langle \sin, \cos \rangle = \int_{-\pi}^{\pi} \sin(x)\cos(x)\, dx = 0.$$

Moreover, $\|\sin\|^2 = \int_{-\pi}^{\pi} \sin^2(x)\, dx = \pi$. More generally, the functions 1, $\cos(x)$, $\sin(x)$, $\cos(2x)$, $\sin(2x)$, … are pairwise orthogonal on $[-\pi, \pi]$. Writing a function in terms of them is the idea behind Fourier series, which decompose a signal into frequencies.

Example 7.3 (Legendre polynomials)**.** Let us apply the Gram–Schmidt process (unnormalized version) to the polynomials $1, x, x^2, x^3$ in $C[-1, 1]$. We need the integrals $\int_{-1}^{1} x^k\, dx$, which are 0 for

odd k and $\frac{2}{k+1}$ for even k:

$$
\begin{aligned}
u_0 &= 1, \\
u_1 &= x - \frac{\langle x, 1\rangle}{\langle 1, 1\rangle} \cdot 1 = x, \\
u_2 &= x^2 - \frac{\langle x^2, 1\rangle}{\langle 1, 1\rangle} \cdot 1 - \frac{\langle x^2, x\rangle}{\langle x, x\rangle}\, x = x^2 - \frac{2/3}{2} = x^2 - \frac{1}{3}, \\
u_3 &= x^3 - \frac{\langle x^3, 1\rangle}{\langle 1, 1\rangle} - \frac{\langle x^3, x\rangle}{\langle x, x\rangle}\, x - \frac{\langle x^3, u_2\rangle}{\langle u_2, u_2\rangle}\, u_2 = x^3 - \frac{2/5}{2/3}\, x = x^3 - \frac{3}{5}x.
\end{aligned}
$$

These orthogonal polynomials, shown in Figure 5, are (up to scaling) the Legendre polynomials. They are used, for example, to approximate functions by polynomials (Exercise 13).

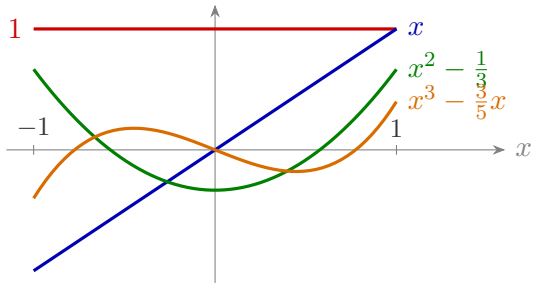


Figure 5: The orthogonal polynomials of Example 7.3 on $[-1, 1]$. Any two of them are orthogonal with respect to $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$.

Remark 7.4 (Random variables). Inner products also appear in probability. For random variables X and Y with mean 0, the covariance $\mathrm{Cov}(X, Y) = \mathbb{E}[XY]$ behaves like an inner product: it is bilinear, symmetric, and $\mathrm{Cov}(X, X) = \mathrm{Var}(X) \geq 0$. The corresponding norm is the standard deviation, and the correlation coefficient $\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$ is the cosine of the angle between X and Y. In particular, uncorrelated random variables are orthogonal, and the Cauchy–Schwarz inequality shows that correlations lie between -1 and 1.

8 Orthogonal Projections

Data in machine learning is often high-dimensional, and it is useful to replace each data point by a point of a lower-dimensional subspace while losing as little information as possible. The best way to do this, in a precise sense, is the orthogonal projection. Orthogonal projections are behind linear regression (Section 8.3), principal component analysis (Section 11.6) and the Gram–Schmidt process.

Definition 8.1 (Orthogonal projection)**.** Let U be a subspace of a finite-dimensional inner product space V. By Theorem 6.5, every $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in U^\perp$. The vector $\pi_U(v) = u$ is called the orthogonal projection of v onto U. In other words, $\pi_U(v)$ is the unique vector with

$$\pi_U(v) \in U \qquad \text{and} \qquad v - \pi_U(v) \in U^\perp.$$

Proposition 8.2. *Let U be a subspace of a finite-dimensional inner product space V.*

1. *If $e_1, \dots, e_k$ is an orthonormal basis of U, then $\pi_U(v) = \sum_{i=1}^{k} \langle v, e_i \rangle e_i$.*

2. *$\pi_U : V \to V$ is a linear transformation with image U and kernel $U^\perp$.*

3. *$\pi_U \circ \pi_U = \pi_U$.*

Proof. (1) This is the formula from the proof of Theorem 6.5. (2) Linearity follows from the formula in part 1. The image is contained in U, and $\pi_U(u) = u$ for $u \in U$ (take $w = 0$), so the image is U. Moreover, $\pi_U(v) = 0$ if and only if $v = 0 + v$ with $v \in U^\perp$. (3) Since $\pi_U(v) \in U$, we get $\pi_U(\pi_U(v)) = \pi_U(v)$. □

A linear transformation π with $\pi \circ \pi = \pi$ is called a projection; for matrices, this is the condition $P^2 = P$ (compare the idempotent matrices in the practice problems of Lecture 1).

8.1 Projection onto a line

Proposition 8.3. *Let $\mathbf{b} \neq \mathbf{0}$ and $U = \operatorname{span}\{\mathbf{b}\}$. Then*

$$\pi_U(\mathbf{x}) = \frac{\langle \mathbf{x}, \mathbf{b} \rangle}{\langle \mathbf{b}, \mathbf{b} \rangle} \mathbf{b}.$$

For the dot product on $\mathbb{R}^n$, this is $\pi_U(\mathbf{x}) = P\mathbf{x}$ with the projection matrix

$$P = \frac{\mathbf{b}\mathbf{b}^\top}{\mathbf{b}^\top \mathbf{b}},$$

which is symmetric and satisfies $P^2 = P$.

Proof. We look for $\pi_U(\mathbf{x}) = \lambda \mathbf{b}$ with $\mathbf{x} - \lambda \mathbf{b} \perp \mathbf{b}$, i.e., $\langle \mathbf{x}, \mathbf{b} \rangle - \lambda \langle \mathbf{b}, \mathbf{b} \rangle = 0$. This gives $\lambda = \frac{\langle \mathbf{x}, \mathbf{b} \rangle}{\langle \mathbf{b}, \mathbf{b} \rangle}$. For the dot product, $\lambda \mathbf{b} = \mathbf{b}\lambda = \frac{\mathbf{b}(\mathbf{b}^\top \mathbf{x})}{\mathbf{b}^\top \mathbf{b}} = P\mathbf{x}$. The matrix $\mathbf{b}\mathbf{b}^\top$ is symmetric, and $P^2 = \frac{\mathbf{b}(\mathbf{b}^\top \mathbf{b})\mathbf{b}^\top}{(\mathbf{b}^\top \mathbf{b})^2} = P$. □

Note that $\mathbf{b}\mathbf{b}^\top$ is an $n \times n$ matrix, while $\mathbf{b}^\top \mathbf{b}$ is a number. The length of the projection is $\|\pi_U(\mathbf{x})\| = \frac{|\langle \mathbf{x}, \mathbf{b} \rangle|}{\|\mathbf{b}\|} = \|\mathbf{x}\| \, |\cos \omega|$, where ω is the angle between $\mathbf{x}$ and $\mathbf{b}$, as in elementary geometry (Figure 6).

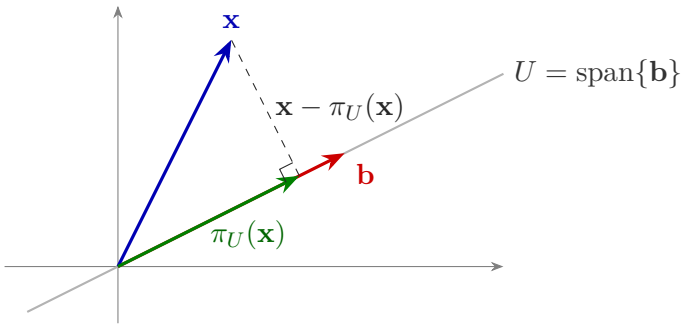


Figure 6: The orthogonal projection of $\mathbf{x} = (1, 2)$ onto the line spanned by $\mathbf{b} = (2, 1)$ is $\pi_U(\mathbf{x}) = \frac{4}{5}\mathbf{b} = (1.6, 0.8)$. The difference $\mathbf{x} - \pi_U(\mathbf{x}) = (-0.6, 1.2)$ is orthogonal to $\mathbf{b}$.

Example 8.4. Let $\mathbf{b} = (1,2,2)^\top$ and $\mathbf{x} = (1,1,1)^\top$. Then $\mathbf{b}^\top\mathbf{b} = 9$ and $\mathbf{b}^\top\mathbf{x} = 5$, so

$$P = \frac{1}{9}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}, \qquad \pi_U(\mathbf{x}) = P\mathbf{x} = \frac{5}{9}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

Indeed, $\mathbf{x} - \pi_U(\mathbf{x}) = \frac{1}{9}(4,-1,-1)^\top$ is orthogonal to $\mathbf{b}$, since $4 - 2 - 2 = 0$.

8.2 Projection onto a subspace

Now let $U = \operatorname{span}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\}$ be an m-dimensional subspace of $\mathbb{R}^n$, with the dot product, and let B be the $n \times m$ matrix with columns $\mathbf{b}_1, \ldots, \mathbf{b}_m$, so that $U = \operatorname{col}(B)$.

Theorem 8.5 (Normal equations)**.** *Let B be an $n \times m$ matrix with linearly independent columns, and $U = \operatorname{col}(B)$. Then for every $\mathbf{x} \in \mathbb{R}^n$,*

$$\pi_U(\mathbf{x}) = B\boldsymbol{\lambda}, \qquad \text{where } \boldsymbol{\lambda} \in \mathbb{R}^m \text{ is the unique solution of } B^\top B\boldsymbol{\lambda} = B^\top\mathbf{x}.$$

In other words, $\boldsymbol{\lambda} = (B^\top B)^{-1}B^\top\mathbf{x}$, and $\pi_U(\mathbf{x}) = P\mathbf{x}$ with the projection matrix

$$P = B(B^\top B)^{-1}B^\top.$$

Proof. Every vector of $U = \operatorname{col}(B)$ has the form $B\boldsymbol{\lambda}$. By Theorem 6.7, $U^\perp = \operatorname{col}(B)^\perp = \operatorname{null}(B^\top)$. So $B\boldsymbol{\lambda} = \pi_U(\mathbf{x})$ if and only if $\mathbf{x} - B\boldsymbol{\lambda} \in \operatorname{null}(B^\top)$, i.e., $B^\top(\mathbf{x} - B\boldsymbol{\lambda}) = \mathbf{0}$, i.e., $B^\top B\boldsymbol{\lambda} = B^\top\mathbf{x}$. The matrix $B^\top B$ is symmetric positive definite by Example 2.9, hence invertible by Proposition 2.7, so this equation has exactly one solution $\boldsymbol{\lambda} = (B^\top B)^{-1}B^\top\mathbf{x}$. □

The equations $B^\top B\boldsymbol{\lambda} = B^\top\mathbf{x}$ are called the normal equations, since they say that $\mathbf{x} - B\boldsymbol{\lambda}$ is orthogonal ("normal") to U. If the columns of B are orthonormal, then $B^\top B = I_m$, and the formulas simplify to $\boldsymbol{\lambda} = B^\top\mathbf{x}$ and $P = BB^\top$, in agreement with Proposition 8.2. For $m = 1$, we recover Proposition 8.3.

Example 8.6. Let $U = \operatorname{span}\{(1,1,1)^\top, (0,1,2)^\top\}$ and $\mathbf{x} = (6,0,0)^\top$. With $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, we get

$$B^\top B = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \qquad B^\top\mathbf{x} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \qquad \boldsymbol{\lambda} = \begin{pmatrix} 5 \\ -3 \end{pmatrix},$$

where $\boldsymbol{\lambda}$ solves the normal equations. Hence

$$\pi_U(\mathbf{x}) = 5\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}, \qquad \mathbf{x} - \pi_U(\mathbf{x}) = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix},$$

and indeed $\mathbf{x} - \pi_U(\mathbf{x})$ is orthogonal to both columns of B. The projection matrix is

$$P = B(B^\top B)^{-1}B^\top = \frac{1}{6}\begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix}.$$

Proposition 8.7. *The projection matrix $P = B(B^\top B)^{-1}B^\top$ is symmetric, and $P^2 = P$.*

Proof. $P^2 = B(B^\top B)^{-1}(B^\top B)(B^\top B)^{-1}B^\top = B(B^\top B)^{-1}B^\top = P$. Since $B^\top B$ is symmetric, so is its inverse (Lecture 1: $(A^{-1})^\top = (A^\top)^{-1}$), and therefore $P^\top = B\big((B^\top B)^{-1}\big)^\top B^\top = P$. □

The most important property of the orthogonal projection is that it gives the closest point.

Theorem 8.8 (Best approximation)**.** *Let U be a subspace of a finite-dimensional inner product space V, and let $v \in V$. Then $\pi_U(v)$ is the point of U that is closest to v:*

$$\|v - \pi_U(v)\| < \|v - u\| \qquad \text{for every } u \in U \text{ with } u \neq \pi_U(v).$$

Proof. Write $v - u = \big(v - \pi_U(v)\big) + \big(\pi_U(v) - u\big)$. The first summand lies in $U^\perp$, and the second one in U, so they are orthogonal. By the theorem of Pythagoras,

$$\|v - u\|^2 = \|v - \pi_U(v)\|^2 + \|\pi_U(v) - u\|^2 > \|v - \pi_U(v)\|^2,$$

since $\pi_U(v) - u \neq 0$. □

The number $\|v - \pi_U(v)\|$ is called the distance from v to U. In Example 8.6, the distance from $\mathbf{x}$ to the plane U is $\|(1, -2, 1)^\top\| = \sqrt{6}$.

Remark 8.9. The Gram–Schmidt process (Theorem 5.6) can now be described in one line: $u_j = v_j - \pi_{U_{j-1}}(v_j)$, where $U_{j-1} = \operatorname{span}\{v_1, \ldots, v_{j-1}\}$. That is, we subtract from v_j its projection onto the span of the previous vectors.

8.3 Least squares and linear regression

Suppose we are given data points $(x_1, y_1), \ldots, (x_N, y_N)$ with $x_i, y_i \in \mathbb{R}$, and we want to find the line $y = w_0 + w_1 x$ that fits the data best. In general, no line passes through all points, so we look for the line that minimizes the sum of squared errors

$$\sum_{i=1}^{N} \big(y_i - (w_0 + w_1 x_i)\big)^2 = \|\mathbf{y} - A\mathbf{w}\|^2, \qquad \text{where} \quad A = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_N \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} w_0 \\ w_1 \end{pmatrix}.$$

The matrix A is called the design matrix. This is a special case of the following problem.

Theorem 8.10 (Least squares)**.** *Let A be an $N \times n$ matrix with linearly independent columns, and let $\mathbf{y} \in \mathbb{R}^N$. Then there is exactly one $\mathbf{w}^* \in \mathbb{R}^n$ that minimizes $\|\mathbf{y} - A\mathbf{w}\|$, namely the solution of the normal equations*

$$A^\top A\, \mathbf{w}^* = A^\top \mathbf{y}, \qquad \textit{i.e.,} \qquad \mathbf{w}^* = (A^\top A)^{-1} A^\top \mathbf{y}.$$

Moreover, $A\mathbf{w}^* = \pi_{\mathrm{col}(A)}(\mathbf{y})$.

Proof. As $\mathbf{w}$ ranges over $\mathbb{R}^n$, the vector $A\mathbf{w}$ ranges over $\mathrm{col}(A)$. By Theorem 8.8, $\|\mathbf{y} - A\mathbf{w}\|$ is minimal exactly when $A\mathbf{w} = \pi_{\mathrm{col}(A)}(\mathbf{y})$, and by Theorem 8.5 (with $B = A$ and $\boldsymbol{\lambda} = \mathbf{w}$), this happens exactly for $\mathbf{w} = (A^\top A)^{-1} A^\top \mathbf{y}$. □

Example 8.11. Let us fit a line to the data points $(-1, 0)$, $(0, 1)$ and $(1, 3)$. The design matrix and the normal equations are

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}, \qquad A^\top A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \quad A^\top \mathbf{y} = \begin{pmatrix} 4 \\ 3 \end{pmatrix},$$

so $\mathbf{w}^* = \left(\frac{4}{3}, \frac{3}{2}\right)^\top$, and the best line is $y = \frac{4}{3} + \frac{3}{2}x$ (Figure 7). The fitted values are $A\mathbf{w}^* = \left(-\frac{1}{6}, \frac{4}{3}, \frac{17}{6}\right)^\top$, the residuals are $\mathbf{y} - A\mathbf{w}^* = \left(\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)^\top$, and the minimal sum of squared errors is $\frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{6}$. As predicted by the theory, the residual vector is orthogonal to both columns of A: its entries sum to 0, and $-\frac{1}{6} + 0 + \frac{1}{6} = 0$.

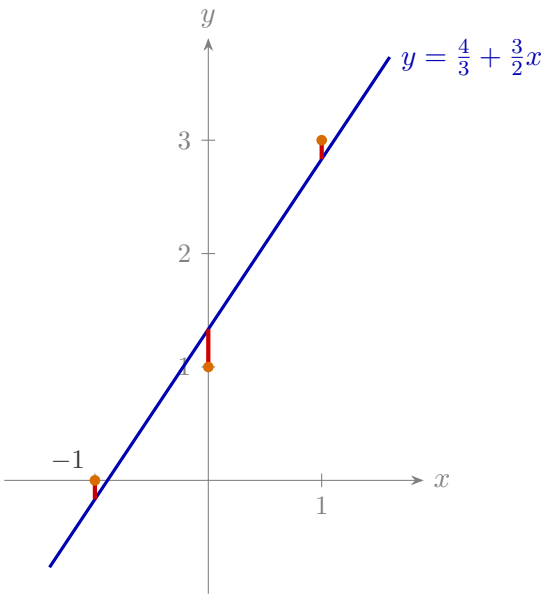


Figure 7: The least squares line for the data points $(-1, 0)$, $(0, 1)$, $(1, 3)$ (Example 8.11). The red segments are the residuals; the line minimizes the sum of their squared lengths.

Remark 8.12 (Linear regression in machine learning). The same method works for any number of features: each row of the design matrix A contains the features of one data point (and a 1 for the intercept w_0), and the least squares solution gives the weights of the linear model. Using the columns $1, x, x^2, \ldots, x^d$, we can also fit polynomials of degree d (polynomial regression). In practice, the normal equations are not solved by computing $(A^\top A)^{-1}$, but with a QR decomposition or with `np.linalg.lstsq`, which are more accurate (compare Lecture 1).

8.4 Projections onto affine subspaces and distances to hyperplanes

Let $L = \mathbf{x}_0 + U$ be an affine subspace, i.e., a subspace U shifted by a vector $\mathbf{x}_0$. To project onto L, we shift everything by $-\mathbf{x}_0$, project onto U, and shift back:

$$\pi_L(\mathbf{x}) = \mathbf{x}_0 + \pi_U(\mathbf{x} - \mathbf{x}_0).$$

Then $\mathbf{x} - \pi_L(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0) - \pi_U(\mathbf{x} - \mathbf{x}_0)$ lies in $U^\perp$, and by Theorem 8.8, $\pi_L(\mathbf{x})$ is the point of L closest to $\mathbf{x}$.

Theorem 8.13 (Distance to a hyperplane)**.** *Let $H = \{\mathbf{z} \in \mathbb{R}^n : \mathbf{w}^\top \mathbf{z} = c\}$ be a hyperplane, where $\mathbf{w} \neq \mathbf{0}$. For every $\mathbf{x} \in \mathbb{R}^n$, the point of H closest to $\mathbf{x}$ is*

$$\pi_H(\mathbf{x}) = \mathbf{x} - \frac{\mathbf{w}^\top \mathbf{x} - c}{\|\mathbf{w}\|^2}\,\mathbf{w}, \qquad \textit{and the distance from } \mathbf{x} \textit{ to } H \textit{ is} \qquad \frac{|\mathbf{w}^\top \mathbf{x} - c|}{\|\mathbf{w}\|}.$$

Proof. Let $\mathbf{x}_0 \in H$, so that $H = \mathbf{x}_0 + W$ with $W = \operatorname{span}\{\mathbf{w}\}^\perp$, and $W^\perp = \operatorname{span}\{\mathbf{w}\}$ by Theorem 6.5(3). Since $\mathbf{v} = \pi_W(\mathbf{v}) + \pi_{W^\perp}(\mathbf{v})$ for every $\mathbf{v}$, we get

$$\mathbf{x} - \pi_H(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0) - \pi_W(\mathbf{x} - \mathbf{x}_0) = \pi_{\operatorname{span}\{\mathbf{w}\}}(\mathbf{x} - \mathbf{x}_0) = \frac{\mathbf{w}^\top(\mathbf{x} - \mathbf{x}_0)}{\mathbf{w}^\top\mathbf{w}}\,\mathbf{w} = \frac{\mathbf{w}^\top\mathbf{x} - c}{\|\mathbf{w}\|^2}\,\mathbf{w},$$

using Proposition 8.3 and $\mathbf{w}^\top\mathbf{x}_0 = c$. Taking the norm gives the distance. □

The number $\frac{\mathbf{w}^\top\mathbf{x}-c}{\|\mathbf{w}\|}$, without the absolute value, is called the signed distance: it is positive on the side of H into which $\mathbf{w}$ points, and negative on the other side.

Example 8.14. The distance from the point $\mathbf{x} = (1, 2, 3)^\top$ to the plane H given by $x + 2y + 2z = 2$ is

$$\frac{|1 + 4 + 6 - 2|}{\sqrt{1 + 4 + 4}} = \frac{9}{3} = 3,$$

and the closest point of H is $\pi_H(\mathbf{x}) = (1, 2, 3)^\top - \frac{9}{9}(1, 2, 2)^\top = (0, 0, 1)^\top$.

Example 8.15. For the linear classifier of Figure 4, with $\mathbf{w} = (1, 1)^\top$ and $b = -2$, the decision boundary is the hyperplane $\mathbf{w}^\top\mathbf{x} = 2$. The signed distance of the point $\mathbf{x} = (2.5, 1.5)^\top$ to it is $\frac{\mathbf{w}^\top\mathbf{x}+b}{\|\mathbf{w}\|} = \frac{4-2}{\sqrt{2}} = \sqrt{2}$, and the closest point on the boundary is $(2.5, 1.5)^\top - \frac{2}{2}(1, 1)^\top = (1.5, 0.5)^\top$.

Remark 8.16 (Margins). For a linear classifier, the distance of a data point to the decision boundary is a measure of confidence: points far from the boundary are classified with a large margin. Support vector machines choose the hyperplane that maximizes the smallest distance from the training points to the boundary, which by Theorem 8.13 means maximizing $\min_i \frac{|\mathbf{w}^\top\mathbf{x}_i+b|}{\|\mathbf{w}\|}$.

9 Rotations

Definition 9.1 (Rotation)**.** A rotation of $\mathbb{R}^n$ is an orthogonal $n \times n$ matrix R with $\det(R) = 1$. The set of rotations is denoted by $\mathrm{SO}(n)$, the special orthogonal group.

By Theorem 4.7, rotations preserve lengths, distances and angles, and the condition $\det(R) = 1$ means that they preserve the orientation (Lecture 1). Orthogonal matrices with determinant -1, such as reflections, reverse the orientation. Since products and inverses of orthogonal matrices are orthogonal, and $\det(R_1R_2) = \det(R_1)\det(R_2)$, products and inverses of rotations are again rotations; so $\mathrm{SO}(n)$ is a matrix group, i.e., a subgroup of $\mathrm{GL}(n, \mathbb{R})$ (Lecture 1). The inverse of a rotation R is simply $R^\top$. Like $\mathrm{O}(n)$, $\mathrm{SO}(n)$ is a Lie group of dimension $\frac{n(n-1)}{2}$ (see the appendix to Lecture 3): a rotation of $\mathbb{R}^3$, for example, is described by three angles (Proposition 9.2).

9.1 Rotations in the plane

In Lecture 1, we met the rotation matrices

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

which rotate the plane counterclockwise by the angle θ. Every rotation R of $\mathbb{R}^2$ is of this form: its first column $(a, b)^\top$ is a unit vector, so $(a, b) = (\cos\theta, \sin\theta)$ for some angle θ; its second column is a unit vector orthogonal to the first, so it is $\pm(-\sin\theta, \cos\theta)^\top$; and $\det(R) = 1$ forces the sign $+$. Since $R_\alpha R_\beta = R_{\alpha+\beta} = R_\beta R_\alpha$, rotations of the plane commute with each other.

9.2 Rotations in three dimensions

The rotations of $\mathbb{R}^3$ by the angle θ about the three coordinate axes are

$$R_1(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \quad R_2(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}, \quad R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For example, $R_3(\theta)$ fixes the x_3-axis and rotates the x_1x_2-plane by θ. Each of them is orthogonal with determinant 1. Every rotation of $\mathbb{R}^3$ is a product of such rotations, and it is a rotation about a line through the origin, its axis.

Proposition 9.2 (Rotations of $\mathbb{R}^3$)**.** *Let $R \in \mathrm{SO}(3)$.*

1. *(Euler angles.) There are angles α, β and γ with $R = R_3(\alpha)R_1(\beta)R_3(\gamma)$.*
2. *(Axis.) There are an orthogonal matrix Q with $\det(Q) = 1$ and an angle θ with $R = QR_3(\theta)Q^\top$. So R fixes the line spanned by the third column $\mathbf{q}_3$ of Q, and it rotates the plane orthogonal to $\mathbf{q}_3$ by the angle θ.*

Proof. We first show that a rotation S with $S\mathbf{e}_3 = \mathbf{e}_3$ is of the form $R_3(\theta)$. Since $S^\top = S^{-1}$, also $S^\top\mathbf{e}_3 = \mathbf{e}_3$, so the third column and the third row of S are $\mathbf{e}_3$ and $\mathbf{e}_3^\top$:

$$S = \begin{pmatrix} S' & 0 \\ 0 & 1 \end{pmatrix}$$

with a 2×2 matrix S'. The columns of S' are orthonormal, since those of S are, and $\det(S') = \det(S) = 1$ (expand along the last row). So S' is a rotation of the plane, $S' = R_\theta$, and $S = R_3(\theta)$.

(1) The vector $\mathbf{w} = R\mathbf{e}_3$ is a unit vector. Choose $\beta \in [0, \pi]$ with $\cos\beta = w_3$. Then $w_1^2 + w_2^2 = 1 - w_3^2 = \sin^2\beta$, so we can choose α with $(w_1, w_2) = (\sin\alpha\sin\beta, -\cos\alpha\sin\beta)$ (any α if $\sin\beta = 0$). A direct computation gives

$$R_3(\alpha)R_1(\beta)\mathbf{e}_3 = R_3(\alpha)(0, -\sin\beta, \cos\beta)^\top = (\sin\alpha\sin\beta, -\cos\alpha\sin\beta, \cos\beta)^\top = \mathbf{w}.$$

Hence $S = \big(R_3(\alpha)R_1(\beta)\big)^{-1}R$ is a rotation with $S\mathbf{e}_3 = \mathbf{e}_3$, so $S = R_3(\gamma)$ for some γ, and $R = R_3(\alpha)R_1(\beta)R_3(\gamma)$.

(2) Since $\det(R) = 1$ and $R^\top R = I_3$,

$$\det(R - I_3) = \det\left(R(I_3 - R^\top)\right) = \det(R)\det\left((I_3 - R)^\top\right) = \det(I_3 - R) = (-1)^3\det(R - I_3).$$

So $\det(R - I_3) = 0$, and 1 is an eigenvalue of R (Lecture 1). Let $\mathbf{q}_3$ be a unit vector with $R\mathbf{q}_3 = \mathbf{q}_3$, and let $\mathbf{q}_1, \mathbf{q}_2$ be an orthonormal basis of $\mathrm{span}\{\mathbf{q}_3\}^\perp$ (Theorem 6.5 and Corollary 5.8). The matrix Q with the columns $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ is orthogonal; if $\det(Q) = -1$, we replace $\mathbf{q}_1$ by $-\mathbf{q}_1$. Then $S = Q^\top RQ$ is a rotation with $S\mathbf{e}_3 = Q^\top R\mathbf{q}_3 = Q^\top\mathbf{q}_3 = \mathbf{e}_3$, so $S = R_3(\theta)$, and $R = QR_3(\theta)Q^\top$. The rotation $R_3(\theta)$ fixes $\mathbf{e}_3$ and rotates the plane spanned by $\mathbf{e}_1$ and $\mathbf{e}_2$, and Q maps $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ to $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$. □

In particular, a rotation of $\mathbb{R}^3$ is determined by its axis and its angle. The Euler angles are used, for example, to describe the orientation of a camera or of a robot arm.

Example 9.3. Unlike in the plane, rotations of $\mathbb{R}^3$ do not commute in general. For $\theta = \frac{\pi}{2}$,

$$R_1(\tfrac{\pi}{2})R_3(\tfrac{\pi}{2}) = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = R_3(\tfrac{\pi}{2})R_1(\tfrac{\pi}{2}).$$

For instance, the first product maps $\mathbf{e}_1$ to $\mathbf{e}_3$, while the second maps $\mathbf{e}_1$ to $\mathbf{e}_2$. You can check this with a book: rotate it first about one axis and then about another, and compare with the opposite order.

9.3 Rotations in n dimensions

Definition 9.4 (Givens rotation)**.** For $1 \le i < j \le n$ and an angle θ, the Givens rotation $G_{ij}(\theta)$ is the $n \times n$ matrix that agrees with the identity matrix, except for the four entries

$$G_{ii} = \cos\theta, \qquad G_{ij} = -\sin\theta, \qquad G_{ji} = \sin\theta, \qquad G_{jj} = \cos\theta.$$

The Givens rotation $G_{ij}(\theta)$ rotates the plane spanned by $\mathbf{e}_i$ and $\mathbf{e}_j$ by the angle θ, and fixes all other standard basis vectors. Its columns are orthonormal, and its determinant is $\cos^2\theta + \sin^2\theta = 1$ (expand along the other rows), so it is a rotation. Givens rotations are used in numerical linear algebra: a well-chosen Givens rotation makes a chosen entry of a matrix zero, and a sequence of them computes a QR decomposition.

Remark 9.5 (Rotations in machine learning)*.* Since rotations preserve dot products and distances, any method that only depends on distances or dot products between data points, such as k-nearest neighbors with the Euclidean distance, gives the same results after the data is rotated. In contrast, methods such as decision trees, which look at individual coordinates, may change. Rotated copies of images are also used for data augmentation, to make image classifiers robust to rotations.

10 The Spectral Theorem

In Lecture 1, we saw that a matrix is diagonalizable if it has a basis of eigenvectors, and that some matrices, such as the shear or a rotation, are not diagonalizable. Symmetric matrices are much better behaved: they always have an orthonormal basis of eigenvectors. Symmetric matrices are everywhere in machine learning: covariance matrices, Gram and kernel matrices, and the matrices $A^\top A$ of least squares problems are all symmetric.

To state the theorem, we use complex eigenvalues (Lecture 1). For a vector $\mathbf{v} \in \mathbb{C}^n$, we write $\bar{\mathbf{v}}$ for the vector of complex conjugates of its entries. Then $\bar{\mathbf{v}}^\top \mathbf{v} = \sum_i \bar{\mathbf{v}}_i v_i = \sum_i |v_i|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$. A complex number λ is a root of p_A if and only if $A\mathbf{v} = \lambda\mathbf{v}$ for some nonzero $\mathbf{v} \in \mathbb{C}^n$, by the same argument as for real eigenvalues.

Theorem 10.1 (Spectral theorem)**.** *Let A be a symmetric real $n \times n$ matrix. Then:*

1. *all complex eigenvalues of A are real;*
2. *eigenvectors of A for distinct eigenvalues are orthogonal;*
3. *there is an orthonormal basis $\mathbf{q}_1, \ldots, \mathbf{q}_n$ of $\mathbb{R}^n$ consisting of eigenvectors of A. Equivalently,*

$$A = Q\Lambda Q^\top,$$

where Q is the orthogonal matrix with columns $\mathbf{q}_1, \ldots, \mathbf{q}_n$, and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ contains the corresponding eigenvalues.

Proof. (1) Let $A\mathbf{v} = \lambda\mathbf{v}$ with $\lambda \in \mathbb{C}$ and $\mathbf{v} \in \mathbb{C}^n$, $\mathbf{v} \neq \mathbf{0}$. The number $z = \bar{\mathbf{v}}^\top A\mathbf{v}$ is real: since A is real and symmetric, its complex conjugate is $\bar{z} = \mathbf{v}^\top A\bar{\mathbf{v}} = (\mathbf{v}^\top A\bar{\mathbf{v}})^\top = \bar{\mathbf{v}}^\top A^\top \mathbf{v} = z$. Since $z = \bar{\mathbf{v}}^\top(\lambda\mathbf{v}) = \lambda\,\bar{\mathbf{v}}^\top\mathbf{v}$, we get $\lambda = z/(\bar{\mathbf{v}}^\top\mathbf{v}) \in \mathbb{R}$.

(2) Let $A\mathbf{v} = \lambda\mathbf{v}$ and $A\mathbf{w} = \mu\mathbf{w}$ with $\lambda \neq \mu$. Then

$$\lambda\,\mathbf{v}^\top\mathbf{w} = (A\mathbf{v})^\top\mathbf{w} = \mathbf{v}^\top A^\top\mathbf{w} = \mathbf{v}^\top A\mathbf{w} = \mu\,\mathbf{v}^\top\mathbf{w},$$

so $(\lambda - \mu)\mathbf{v}^\top\mathbf{w} = 0$, and hence $\mathbf{v}^\top\mathbf{w} = 0$.

(3) We use induction on n; the case $n = 1$ is clear. By part 1 and the fundamental theorem of algebra, p_A has a real root λ_1. Let $\mathbf{q}_1$ be a unit eigenvector for λ_1 (it can be chosen real, since $A - \lambda_1 I_n$ is a real matrix with a nonzero null space). Let $W = \operatorname{span}\{\mathbf{q}_1\}^\perp$, and let $\mathbf{w}_2, \dots, \mathbf{w}_n$ be an orthonormal basis of W (Theorem 6.5 and Corollary 5.8). The subspace W is mapped into itself by A: if $\mathbf{x} \in W$, then $\mathbf{q}_1^\top (A\mathbf{x}) = (A\mathbf{q}_1)^\top \mathbf{x} = \lambda_1 \mathbf{q}_1^\top \mathbf{x} = 0$.

Let M be the $n \times (n-1)$ matrix with columns $\mathbf{w}_2, \dots, \mathbf{w}_n$, so that $M^\top M = I_{n-1}$, and let $B = M^\top A M$, which is a symmetric $(n-1) \times (n-1)$ matrix. Since $A\mathbf{w}_i \in W$, Proposition 5.3 gives $A\mathbf{w}_i = \sum_k (\mathbf{w}_k^\top A \mathbf{w}_i)\mathbf{w}_k$, i.e., $AM = MB$. By induction, B has an orthonormal basis of eigenvectors $\mathbf{y}_2, \dots, \mathbf{y}_n \in \mathbb{R}^{n-1}$, with $B\mathbf{y}_j = \lambda_j \mathbf{y}_j$. Then the vectors $\mathbf{q}_j = M\mathbf{y}_j$ $(j = 2, \dots, n)$ are orthonormal, since $\mathbf{q}_i^\top \mathbf{q}_j = \mathbf{y}_i^\top M^\top M \mathbf{y}_j = \mathbf{y}_i^\top \mathbf{y}_j$; they lie in W, so they are orthogonal to $\mathbf{q}_1$; and they are eigenvectors, since $A\mathbf{q}_j = AM\mathbf{y}_j = MB\mathbf{y}_j = \lambda_j \mathbf{q}_j$. So $\mathbf{q}_1, \dots, \mathbf{q}_n$ is an orthonormal basis of eigenvectors.

Finally, by Theorem 4.7, Q is orthogonal, so $Q^{-1} = Q^\top$, and by the diagonalization theorem of Lecture 1, $A = Q\Lambda Q^{-1} = Q\Lambda Q^\top$. □

The factorization $A = Q\Lambda Q^\top$ is called the eigendecomposition, or spectral decomposition, of A. Multiplying out, it says that

$$A = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^\top + \lambda_2 \mathbf{q}_2 \mathbf{q}_2^\top + \cdots + \lambda_n \mathbf{q}_n \mathbf{q}_n^\top,$$

where each $\mathbf{q}_i \mathbf{q}_i^\top$ is the matrix of the orthogonal projection onto the line $\operatorname{span}\{\mathbf{q}_i\}$ (Proposition 8.3). So a symmetric matrix projects a vector onto n orthogonal directions, scales the projections by the eigenvalues, and adds them up.

Example 10.2. Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = J + 2I_3$, where $J = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ is the matrix from the section on eigenvalues in Lecture 1. By the computations there, the eigenvalues of A are $2 + 2 = 4$, with eigenspace $\operatorname{span}\{(1,1,1)^\top\}$, and $-1 + 2 = 1$, with eigenspace the plane $x + y + z = 0$. As predicted by the spectral theorem, the two eigenspaces are orthogonal. To get an orthonormal basis of the plane, we apply the Gram–Schmidt process to its basis $(1,-1,0)^\top$, $(1,0,-1)^\top$, which gives $\frac{1}{\sqrt{2}}(1,-1,0)^\top$ and $\frac{1}{\sqrt{6}}(1,1,-2)^\top$. Hence $A = Q\Lambda Q^\top$ with

$$Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(4,1,1).$$

Corollary 10.3. *A symmetric matrix is positive definite if and only if all its eigenvalues are positive, and positive semidefinite if and only if all its eigenvalues are nonnegative.*

Proof. Write $A = Q\Lambda Q^\top$ and $\mathbf{y} = Q^\top \mathbf{x}$. Then

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{y}^\top \Lambda \mathbf{y} = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.$$

If all $\lambda_i > 0$, this is positive for $\mathbf{x} \neq \mathbf{0}$ (then $\mathbf{y} \neq \mathbf{0}$, since $Q^\top$ is invertible). Conversely, $\mathbf{q}_i^\top A \mathbf{q}_i = \lambda_i$, so if A is positive definite, all $\lambda_i > 0$. The semidefinite case is the same, with $\geq$ instead of $>$. □

Corollary 10.4 (Sylvester's criterion)**.** *Let A be a symmetric $n \times n$ matrix, and let A_k be its upper left $k \times k$ submatrix, with the entries A_{ij} for $i, j \leq k$. Then A is positive definite if and only if* $\det(A_k) > 0$ *for $k = 1, \dots, n$.*

Proof. For $\mathbf{x} \in \mathbb{R}^k$, let $\tilde{\mathbf{x}} \in \mathbb{R}^n$ be $\mathbf{x}$ followed by $n-k$ zeros. Then $\mathbf{x}^\top A_k \mathbf{x} = \tilde{\mathbf{x}}^\top A \tilde{\mathbf{x}}$. So if A is positive definite, then so is every A_k. By Corollary 10.3, the eigenvalues of A_k are positive, and $\det(A_k)$, which is their product (Lecture 1), is positive.

For the converse, we use induction on n; for $n = 1$, $A = (A_{11})$ with $A_{11} > 0$. Let $n \geq 2$, and assume that all $\det(A_k) > 0$. By induction, A_{n-1} is positive definite. The eigenvalues $\lambda_1, \ldots, \lambda_n$ of A have the positive product $\det(A)$, so none of them is 0, and the number of negative ones is even. Suppose that A is not positive definite. Then at least two eigenvalues are negative, say $\lambda_i, \lambda_j < 0$ with $i \neq j$, with orthonormal eigenvectors $\mathbf{q}_i$ and $\mathbf{q}_j$. The equation $a(\mathbf{q}_i)_n + b(\mathbf{q}_j)_n = 0$ has a solution $(a, b) \neq (0, 0)$, and then $\mathbf{x} = a\mathbf{q}_i + b\mathbf{q}_j$ is nonzero (as $\mathbf{q}_i$ and $\mathbf{q}_j$ are linearly independent), and its last entry is 0. On the one hand, $\mathbf{x}^\top A \mathbf{x} = \lambda_i a^2 + \lambda_j b^2 < 0$. On the other hand, $\mathbf{x} = \tilde{\mathbf{x}'}$ for some nonzero $\mathbf{x}' \in \mathbb{R}^{n-1}$, so $\mathbf{x}^\top A \mathbf{x} = \mathbf{x}'^\top A_{n-1} \mathbf{x}' > 0$. This is a contradiction. □

For example, the matrix $A = J + 2I_3$ of Example 10.2 has $\det(A_1) = 2$, $\det(A_2) = 3$ and $\det(A_3) = 4$, so it is positive definite, in agreement with its eigenvalues $4, 1, 1$.

Example 10.5. The matrix $\begin{pmatrix} 9 & 6 \\ 6 & 5 \end{pmatrix}$ from the subsection on symmetric positive definite matrices has trace 14 and determinant 9, so its eigenvalues $7 \pm 2\sqrt{10}$ are both positive, and it is positive definite. The matrix $\begin{pmatrix} 9 & 6 \\ 6 & 3 \end{pmatrix}$ has determinant $-9 < 0$, so its eigenvalues have opposite signs (their product is negative), and it is not positive definite.

Positive semidefinite matrices have square roots, just like nonnegative numbers.

Proposition 10.6 (Square root)**.** *Let $A = Q\Lambda Q^\top$ be symmetric positive semidefinite, with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then $A^{1/2} = Q\Lambda^{1/2}Q^\top$, where $\Lambda^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$, is the only symmetric positive semidefinite matrix B with $B^2 = A$. If A is positive definite, then so is $A^{1/2}$, and $A^{-1/2} = (A^{1/2})^{-1} = Q\Lambda^{-1/2}Q^\top$.*

Proof. The matrix $Q\Lambda^{1/2}Q^\top$ is symmetric, its eigenvalues $\sqrt{\lambda_i}$ are nonnegative, and $(Q\Lambda^{1/2}Q^\top)^2 = Q\Lambda^{1/2}Q^\top Q\Lambda^{1/2}Q^\top = Q\Lambda Q^\top = A$. For uniqueness, let B be symmetric positive semidefinite with $B^2 = A$, and write $B = PMP^\top$ with $M = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, all $\mu_i \geq 0$. Then $A = PM^2P^\top$, so the columns $\mathbf{p}_i$ of P are eigenvectors of A with the eigenvalues μ_i^2. For an eigenvalue λ of A, the eigenspace E_λ of A is spanned by the $\mathbf{p}_i$ with $\mu_i^2 = \lambda$ (Lecture 1, proof of the theorem on multiplicities), i.e., with $\mu_i = \sqrt{\lambda}$, since $\mu_i \geq 0$. So $B\mathbf{v} = \sqrt{\lambda}\,\mathbf{v}$ for every $\mathbf{v} \in E_\lambda$. The eigenspaces of A together span $\mathbb{R}^n$, so B is determined by A. If all $\lambda_i > 0$, then all $\sqrt{\lambda_i} > 0$, and $Q\Lambda^{-1/2}Q^\top$ is the inverse of $Q\Lambda^{1/2}Q^\top$. □

For example, $\begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ has the eigenvalues 9 and 1, with eigenvectors $(1, 1)^\top$ and $(1, -1)^\top$, and its square root is $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with eigenvalues 3 and 1. If Σ is the covariance matrix of data, multiplying the centered data by $\Sigma^{-1/2}$ gives data with the covariance matrix I (whitening; see Lecture 4). Both $A = A^{1/2}A^{1/2}$ and the Cholesky decomposition $A = LL^\top$ write A as $BB^\top$; they differ by an orthogonal matrix, since $L^{-1}A^{1/2}$ is orthogonal (Exercise 19).

Proposition 10.7 (Rayleigh quotient)**.** *Let A be symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ and orthonormal eigenvectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$. Then for every $\mathbf{x} \neq \mathbf{0}$,*

$$\lambda_n \leq \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_1,$$

with equality on the right for $\mathbf{x} = \mathbf{q}_1$*, and on the left for* $\mathbf{x} = \mathbf{q}_n$*.*

Proof. With $\mathbf{y} = Q^\top \mathbf{x}$, we have $\mathbf{x}^\top \mathbf{x} = \mathbf{y}^\top \mathbf{y}$ (Theorem 4.7) and $\mathbf{x}^\top A \mathbf{x} = \sum_i \lambda_i y_i^2$, which lies between $\lambda_n \sum_i y_i^2$ and $\lambda_1 \sum_i y_i^2$. For $\mathbf{x} = \mathbf{q}_i$, the quotient is $\mathbf{q}_i^\top A \mathbf{q}_i = \lambda_i$. □

So the largest eigenvalue of a symmetric matrix is the maximum of $\mathbf{x}^\top A \mathbf{x}$ over all unit vectors $\mathbf{x}$, and it is attained at an eigenvector. This is the key to principal component analysis. More generally, the sum of the k largest eigenvalues is the maximum of $\sum_j \mathbf{b}_j^\top A \mathbf{b}_j$ over k orthonormal vectors $\mathbf{b}_1, \dots, \mathbf{b}_k$.

Proposition 10.8 (Ky Fan)**.** *Let* A *be symmetric with eigenvalues* $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ *and orthonormal eigenvectors* $\mathbf{q}_1, \dots, \mathbf{q}_n$*, and let* $1 \leq k \leq n$*. For every* $n \times k$ *matrix* B *with orthonormal columns* $\mathbf{b}_1, \dots, \mathbf{b}_k$*,*

$$\operatorname{tr}(B^\top A B) = \sum_{j=1}^{k} \mathbf{b}_j^\top A \mathbf{b}_j \leq \lambda_1 + \cdots + \lambda_k,$$

with equality for $\mathbf{b}_j = \mathbf{q}_j$*,* $j = 1, \dots, k$*.*

Proof. Writing $\mathbf{b}_j = \sum_i (\mathbf{q}_i^\top \mathbf{b}_j) \mathbf{q}_i$ (Proposition 5.3) gives $\mathbf{b}_j^\top A \mathbf{b}_j = \sum_i \lambda_i (\mathbf{q}_i^\top \mathbf{b}_j)^2$, so

$$\operatorname{tr}(B^\top A B) = \sum_{i=1}^{n} \lambda_i c_i, \qquad \text{where } c_i = \sum_{j=1}^{k} (\mathbf{q}_i^\top \mathbf{b}_j)^2 = \|B^\top \mathbf{q}_i\|^2.$$

The numbers c_i satisfy $\sum_i c_i = \sum_j \sum_i (\mathbf{q}_i^\top \mathbf{b}_j)^2 = \sum_j \|\mathbf{b}_j\|^2 = k$ and $0 \leq c_i \leq 1$: extending $\mathbf{b}_1, \dots, \mathbf{b}_k$ to an orthonormal basis $\mathbf{b}_1, \dots, \mathbf{b}_n$ of $\mathbb{R}^n$ gives $c_i \leq \sum_{j=1}^{n} (\mathbf{b}_j^\top \mathbf{q}_i)^2 = \|\mathbf{q}_i\|^2 = 1$. Since $\lambda_i \geq \lambda_k$ and $c_i - 1 \leq 0$ for $i \leq k$, and $\lambda_i \leq \lambda_k$ and $c_i \geq 0$ for $i > k$,

$$\sum_{i=1}^{n} \lambda_i c_i - \sum_{i=1}^{k} \lambda_i = \sum_{i \leq k} \lambda_i (c_i - 1) + \sum_{i > k} \lambda_i c_i \leq \lambda_k \sum_{i \leq k} (c_i - 1) + \lambda_k \sum_{i > k} c_i = \lambda_k \Big(\sum_{i=1}^{n} c_i - k \Big) = 0.$$

For $\mathbf{b}_j = \mathbf{q}_j$, $\operatorname{tr}(B^\top A B) = \sum_{j \leq k} \mathbf{q}_j^\top A \mathbf{q}_j = \lambda_1 + \cdots + \lambda_k$. □

The Rayleigh quotient $R_A(\mathbf{x}) = \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$ also describes the eigenvalues between λ_1 and λ_n.

Theorem 10.9 (Courant–Fischer)**.** *Let* A *be symmetric with eigenvalues* $\lambda_1 \geq \cdots \geq \lambda_n$*. For* $k = 1, \dots, n$*,*

$$\lambda_k = \max_{\dim U = k} \; \min_{\mathbf{x} \in U, \, \mathbf{x} \neq \mathbf{0}} R_A(\mathbf{x}),$$

where the maximum is over all k*-dimensional subspaces* U *of* $\mathbb{R}^n$*, and it is attained for* $U = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$*.*

Proof. Write $\mathbf{x} = \sum_i y_i \mathbf{q}_i$, so that $R_A(\mathbf{x}) = \sum_i \lambda_i y_i^2 / \sum_i y_i^2$. For $U = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$, every nonzero $\mathbf{x} \in U$ has $y_i = 0$ for $i > k$, so $R_A(\mathbf{x}) \geq \lambda_k$, with equality for $\mathbf{x} = \mathbf{q}_k$; so the minimum over this U is λ_k. Now let U be any k-dimensional subspace. The subspace $W = \operatorname{span}\{\mathbf{q}_k, \dots, \mathbf{q}_n\}$ has dimension $n - k + 1$, and $\dim U + \dim W > n$, so U and W contain a common nonzero vector $\mathbf{x}$, as in the proof of Theorem 11.7. It has $y_i = 0$ for $i < k$, so $R_A(\mathbf{x}) \leq \lambda_k$. Hence the minimum over U is at most λ_k. □

Corollary 10.10 (Weyl)**.** *Let* A *and* E *be symmetric, and let all eigenvalues of* E *lie in* $[-\varepsilon, \varepsilon]$*. Then the* k*-th largest eigenvalues of* A *and* $A + E$ *differ by at most* ε*:* $|\lambda_k(A + E) - \lambda_k(A)| \leq \varepsilon$ *for every* k*.*

Proof. By Proposition 10.7, $-\varepsilon \le R_E(\mathbf{x}) \le \varepsilon$, so $R_A(\mathbf{x}) - \varepsilon \le R_{A+E}(\mathbf{x}) = R_A(\mathbf{x}) + R_E(\mathbf{x}) \le R_A(\mathbf{x}) + \varepsilon$ for all $\mathbf{x} \neq \mathbf{0}$. Taking the minimum over U and then the maximum over all k-dimensional U preserves these inequalities, and Theorem 10.9 gives $\lambda_k(A) - \varepsilon \le \lambda_k(A+E) \le \lambda_k(A) + \varepsilon$. □

So the eigenvalues of a symmetric matrix depend continuously, and even with Lipschitz constant 1, on its entries. For example, the eigenvalues of a covariance matrix that is estimated from data are close to the true ones when the estimate is close to the true matrix, which justifies PCA on estimated covariance matrices. For nonsymmetric matrices, this fails badly: $\begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}$ has the eigenvalues $\pm\sqrt{\delta}$, which change by 10^{-3} when the entry δ changes from 0 to 10^{-6}.

Remark 10.11 (Geometry of symmetric positive definite matrices). If A is symmetric positive definite, the set $\{\mathbf{x} : \mathbf{x}^\top A\mathbf{x} = 1\}$ is an ellipse (in $\mathbb{R}^2$) or an ellipsoid (in higher dimensions), whose axes point in the directions of the eigenvectors $\mathbf{q}_i$ and have lengths $\frac{1}{\sqrt{\lambda_i}}$, since in the coordinates $\mathbf{y} = Q^\top\mathbf{x}$ the equation becomes $\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = 1$. In machine learning, such ellipsoids describe the shape of Gaussian distributions (with A the inverse covariance matrix) and the shape of the loss surface of least squares problems near a minimum; when the eigenvalues have very different sizes, gradient descent zig-zags and converges slowly.

10.1 The graph Laplacian and spectral clustering

Many data sets are graphs: users connected by friendships, web pages by links, molecules by bonds, or data points connected to their nearest neighbors. A weighted graph on the vertices $1, \dots, n$ is described by a symmetric $n \times n$ matrix W of weights $w_{ij} = w_{ji} \ge 0$, with $w_{ii} = 0$; there is an edge between i and j if $w_{ij} > 0$. For example, $w_{ij} = 1$ for the pairs of friends, or $w_{ij} = e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2/(2s^2)}$ for data points $\mathbf{x}_i$ that are close to each other. A symmetric matrix describes the graph, and the spectral theorem describes its shape.

Definition 10.12 (Graph Laplacian)**.** The degree of the vertex i is $d_i = \sum_j w_{ij}$. The Laplacian of the graph is the matrix $L = D - W$, where $D = \operatorname{diag}(d_1, \dots, d_n)$.

Proposition 10.13. *For every $\mathbf{x} \in \mathbb{R}^n$,*

$$\mathbf{x}^\top L\mathbf{x} = \sum_{i<j} w_{ij}(x_i - x_j)^2.$$

Hence L is symmetric and positive semidefinite, and $L\mathbf{1} = \mathbf{0}$ for the vector $\mathbf{1} = (1, \dots, 1)^\top$.

Proof. $\mathbf{x}^\top L\mathbf{x} = \sum_i d_i x_i^2 - \sum_{i,j} w_{ij} x_i x_j = \frac{1}{2}\sum_{i,j} w_{ij}(x_i^2 + x_j^2 - 2x_i x_j) = \frac{1}{2}\sum_{i,j} w_{ij}(x_i - x_j)^2$, using $\sum_i d_i x_i^2 = \sum_{i,j} w_{ij} x_i^2 = \sum_{i,j} w_{ij} x_j^2$; the terms with $i = j$ vanish, and those with $i > j$ repeat those with $i < j$. The rows of L sum to $d_i - \sum_j w_{ij} = 0$. □

So $\mathbf{x}^\top L\mathbf{x}$ is small if $\mathbf{x}$ varies little along the edges with large weights: it measures how smooth a function $\mathbf{x}$ on the vertices is. The vertices i and j are in the same connected component of the graph if there is a path $i = i_0, i_1, \dots, i_l = j$ with $w_{i_{s-1} i_s} > 0$ for every step.

Theorem 10.14. *Let L be the Laplacian of a graph with the connected components $C_1, \dots, C_k$. Then* $\operatorname{null}(L)$ *is spanned by the indicator vectors $\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_k}$ (with the entries* 1 *on C_l and* 0 *elsewhere). So the multiplicity of the eigenvalue* 0 *is the number of connected components; in particular, the graph is connected if and only if the second smallest eigenvalue λ_2 of L is positive.*

Proof. Write $L = Q\Lambda Q^\top$ with eigenvalues $\lambda_i \geq 0$ (Corollary 10.3), and let $\mathbf{y} = Q^\top \mathbf{x}$. If $\mathbf{x}^\top L\mathbf{x} = \sum_i \lambda_i y_i^2 = 0$, then $y_i = 0$ whenever $\lambda_i > 0$, so $L\mathbf{x} = Q\Lambda\mathbf{y} = \mathbf{0}$; the converse is clear. So $L\mathbf{x} = \mathbf{0}$ if and only if $\mathbf{x}^\top L\mathbf{x} = 0$, i.e., by Proposition 10.13, if and only if $x_i = x_j$ whenever $w_{ij} > 0$. Following paths, this means that $\mathbf{x}$ is constant on each connected component, i.e., $\mathbf{x} = \sum_l c_l \mathbf{1}_{C_l}$. The vectors $\mathbf{1}_{C_l}$ are nonzero and orthogonal, hence linearly independent (Lemma 5.2), so $\dim \operatorname{null}(L) = k$, which is the multiplicity of the eigenvalue 0 in the spectral decomposition. □

Spectral clustering. Clustering means splitting the vertices into groups with many edges inside the groups and few edges between them. For a set A of vertices and its complement $\bar{A}$, the weight of the edges between them is $\operatorname{cut}(A) = \sum_{i\in A,\, j\in\bar{A}} w_{ij}$. Minimizing the cut alone tends to split off single vertices, so one balances it by the sizes of the parts:

$$\operatorname{RatioCut}(A) = \operatorname{cut}(A)\Big(\frac{1}{|A|} + \frac{1}{|\bar{A}|}\Big).$$

Finding the best A is a hard combinatorial problem, since there are $2^{n-1} - 1$ ways to split the vertices into two nonempty parts. The Laplacian turns it into an eigenvalue problem.

Theorem 10.15. *Let $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ be the eigenvalues of the Laplacian of a graph. For every set of vertices A with $A \neq \emptyset$ and $\bar{A} \neq \emptyset$, define $\mathbf{f} \in \mathbb{R}^n$ by $f_i = \sqrt{|\bar{A}|/|A|}$ for $i \in A$ and $f_i = -\sqrt{|A|/|\bar{A}|}$ for $i \in \bar{A}$. Then*

$$\mathbf{f}^\top\mathbf{1} = 0, \qquad \|\mathbf{f}\|^2 = n, \qquad \mathbf{f}^\top L\mathbf{f} = n\operatorname{RatioCut}(A), \qquad \textit{and hence} \qquad \operatorname{RatioCut}(A) \geq \lambda_2.$$

Proof. $\mathbf{f}^\top\mathbf{1} = |A|\sqrt{|\bar{A}|/|A|} - |\bar{A}|\sqrt{|A|/|\bar{A}|} = \sqrt{|A||\bar{A}|} - \sqrt{|A||\bar{A}|} = 0$, and $\|\mathbf{f}\|^2 = |\bar{A}| + |A| = n$. In Proposition 10.13, only the edges between A and $\bar{A}$ contribute, each with

$$(f_i - f_j)^2 = \frac{|\bar{A}|}{|A|} + \frac{|A|}{|\bar{A}|} + 2 = \frac{(|A| + |\bar{A}|)^2}{|A||\bar{A}|} = n\Big(\frac{1}{|A|} + \frac{1}{|\bar{A}|}\Big),$$

which gives $\mathbf{f}^\top L\mathbf{f} = n\operatorname{RatioCut}(A)$. Let $\mathbf{q}_1 = \frac{1}{\sqrt{n}}\mathbf{1}, \mathbf{q}_2, \ldots, \mathbf{q}_n$ be orthonormal eigenvectors of L. Since $\mathbf{f}$ is orthogonal to $\mathbf{q}_1$, it is $\mathbf{f} = \sum_{i\geq 2} c_i\mathbf{q}_i$, and $\mathbf{f}^\top L\mathbf{f} = \sum_{i\geq 2}\lambda_i c_i^2 \geq \lambda_2\|\mathbf{f}\|^2 = n\lambda_2$. □

The proof also shows that λ_2 is the minimum of $\frac{\mathbf{x}^\top L\mathbf{x}}{\mathbf{x}^\top\mathbf{x}}$ over all $\mathbf{x} \neq \mathbf{0}$ with $\mathbf{x}^\top\mathbf{1} = 0$, attained at an eigenvector $\mathbf{q}_2$, which is called the Fiedler vector. Minimizing over the special vectors $\mathbf{f}$ of the theorem is hard, but minimizing over all vectors orthogonal to $\mathbf{1}$ is an eigenvalue problem. This suggests spectral clustering: compute the Fiedler vector $\mathbf{q}_2$, and put the vertices with $(\mathbf{q}_2)_i \geq 0$ into one cluster and the others into the second one. For k clusters, one represents each vertex i by the i-th row of the $n \times k$ matrix of the eigenvectors $\mathbf{q}_1, \ldots, \mathbf{q}_k$ and clusters these points with k-means (Lecture 4).

Example 10.16. Take two triangles on the vertices $\{1, 2, 3\}$ and $\{4, 5, 6\}$, joined by the edge between 3 and 4, all with weight 1. One checks that $L\mathbf{x} = \lambda\mathbf{x}$ for $\mathbf{x} = (1, 1, t, -t, -1, -1)^\top$ if $1 - t = \lambda$ (the equation for vertex 1) and $4t - 2 = \lambda t$ (vertex 3), i.e., $\lambda^2 - 5\lambda + 2 = 0$. The smaller root $\lambda_2 = \frac{5-\sqrt{17}}{2} \approx 0.44$ is the second smallest eigenvalue (the others are 0, 3, 3, 3 and $\frac{5+\sqrt{17}}{2}$), with $t = \frac{\sqrt{17}-3}{2} \approx 0.56$. The signs of the Fiedler vector separate the two triangles, and for $A = \{1, 2, 3\}$, $\operatorname{RatioCut}(A) = 1 \cdot (\frac{1}{3} + \frac{1}{3}) = \frac{2}{3} \geq \lambda_2$, as predicted.

Remark 10.17 (Graphs in machine learning). The quadratic form $\mathbf{x}^\top L\mathbf{x}$ is used as a smoothness penalty in semi-supervised learning: labels known on a few vertices are extended to the whole graph by minimizing $\mathbf{x}^\top L\mathbf{x}$ subject to the known values, so that neighbors get similar labels. Laplacian eigenmaps embed data points by the eigenvectors $\mathbf{q}_2, \mathbf{q}_3, \ldots$, and graph neural networks update the features of each vertex by averaging over its neighbors with matrices such as $D^{-1/2}WD^{-1/2}$, which are closely related to the Laplacian.

11 The Singular Value Decomposition

The eigendecomposition only exists for square matrices, and in general only for diagonalizable ones. The singular value decomposition (SVD) exists for every matrix, square or not. It is one of the most useful factorizations in applied mathematics and machine learning: it describes the geometry of every linear transformation, gives the best low-rank approximations of a matrix, and is the basis of principal component analysis.

11.1 Existence of the SVD

Theorem 11.1 (Singular value decomposition)**.** *Let A be a real $m \times n$ matrix. Then there are an orthogonal $m \times m$ matrix U, an orthogonal $n \times n$ matrix V, and an $m \times n$ matrix Σ with*

$$\Sigma_{11} = \sigma_1 \geq \Sigma_{22} = \sigma_2 \geq \cdots \geq \Sigma_{rr} = \sigma_r > 0$$

and all other entries 0*, such that*

$$A = U\Sigma V^\top.$$

The number r of nonzero diagonal entries is the rank of A. The numbers $\sigma_1, \ldots, \sigma_r$ are called the singular values of A; the columns $\mathbf{u}_1, \ldots, \mathbf{u}_m$ of U and $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of V are called the left and right singular vectors of A.

Proof. The $n \times n$ matrix $A^\top A$ is symmetric, and positive semidefinite, since $\mathbf{x}^\top A^\top A\mathbf{x} = \|A\mathbf{x}\|^2 \geq 0$. By the spectral theorem and Corollary 10.3, there is an orthonormal basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of $\mathbb{R}^n$ of eigenvectors of $A^\top A$, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ (we order them by size). Note that

$$\|A\mathbf{v}_i\|^2 = \mathbf{v}_i^\top A^\top A\mathbf{v}_i = \lambda_i\, \mathbf{v}_i^\top \mathbf{v}_i = \lambda_i.$$

Let r be the number of positive eigenvalues, and let $\sigma_i = \sqrt{\lambda_i}$ and $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ for $i = 1, \ldots, r$. These vectors are orthonormal:

$$\mathbf{u}_i^\top \mathbf{u}_j = \frac{\mathbf{v}_i^\top A^\top A\mathbf{v}_j}{\sigma_i\sigma_j} = \frac{\lambda_j\, \mathbf{v}_i^\top \mathbf{v}_j}{\sigma_i\sigma_j} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Let $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$ be an orthonormal basis of $\operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}^\perp$ (Theorem 6.5 and Corollary 5.8), so that $\mathbf{u}_1, \ldots, \mathbf{u}_m$ is an orthonormal basis of $\mathbb{R}^m$, and let U and V be the orthogonal matrices with these columns. For $i \leq r$, we have $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ by definition, and for $i > r$, we have $\|A\mathbf{v}_i\|^2 = \lambda_i = 0$, so $A\mathbf{v}_i = \mathbf{0}$. Column by column, this says that $AV = U\Sigma$, and hence $A = U\Sigma V^{-1} = U\Sigma V^\top$.

Finally, every vector $A\mathbf{x} = \sum_i (\mathbf{v}_i^\top \mathbf{x}) A\mathbf{v}_i = \sum_{i \leq r} \sigma_i (\mathbf{v}_i^\top \mathbf{x}) \mathbf{u}_i$ lies in $\operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$, and conversely $\mathbf{u}_i = A(\frac{1}{\sigma_i}\mathbf{v}_i) \in \operatorname{col}(A)$. So $\operatorname{col}(A) = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$, and since $\mathbf{u}_1, \ldots, \mathbf{u}_r$ are linearly independent (Lemma 5.2), $\operatorname{rank}(A) = r$. □

The proof shows how to compute the SVD by hand:

1. compute the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ and orthonormal eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of $A^\top A$;
2. set $\sigma_i = \sqrt{\lambda_i}$ and $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ for the positive eigenvalues;
3. complete $\mathbf{u}_1, \ldots, \mathbf{u}_r$ to an orthonormal basis of $\mathbb{R}^m$.

Similarly, $AA^\top = U\Sigma\Sigma^\top U^\top$, so the left singular vectors are eigenvectors of $AA^\top$, with the same nonzero eigenvalues σ_i^2.

Example 11.2. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then $A^\top A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, which we diagonalized in Lecture 1: its eigenvalues are 3 and 1, with orthonormal eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. So $\sigma_1 = \sqrt{3}$ and $\sigma_2 = 1$, and

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} A\mathbf{v}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = \frac{1}{1} A\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

A unit vector orthogonal to both is $\mathbf{u}_3 = \frac{1}{\sqrt{3}}(1,-1,-1)^\top$. Hence

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{V^\top}.$$

11.2 The geometry of the SVD

Since U and V are orthogonal, the SVD says that every linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is a composition of three simple steps: an orthogonal transformation $V^\top$ (a rotation, possibly combined with a reflection), a scaling Σ of the coordinate axes by the singular values (which may also change the dimension), and another orthogonal transformation U. In particular, A maps the unit sphere of $\mathbb{R}^n$ to an ellipsoid (possibly flattened) in $\mathbb{R}^m$, whose semi-axes are $\sigma_1\mathbf{u}_1, \ldots, \sigma_r\mathbf{u}_r$: the right singular vector v_i is mapped to $\sigma_i\mathbf{u}_i$.

Example 11.3 (The SVD of a shear)**.** Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ be the shear from the gallery of linear transformations in Lecture 1. Then $A^\top A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has trace 3 and determinant 1, so its eigenvalues are $\frac{3 \pm \sqrt{5}}{2} = \varphi^2$ and φ^{-2}, where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio, which we met in Lecture 1 in Binet's formula for the Fibonacci numbers. So the singular values of the shear are

$$\sigma_1 = \varphi \approx 1.618 \qquad \text{and} \qquad \sigma_2 = \frac{1}{\varphi} \approx 0.618.$$

Using $\varphi^2 = \varphi + 1$, one checks that, with $c = \sqrt{1+\varphi^2}$,

$$\mathbf{v}_1 = \frac{1}{c} \begin{pmatrix} 1 \\ \varphi \end{pmatrix}, \quad \mathbf{v}_2 = \frac{1}{c} \begin{pmatrix} -\varphi \\ 1 \end{pmatrix}, \quad \mathbf{u}_1 = \frac{1}{c} \begin{pmatrix} \varphi \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \frac{1}{c} \begin{pmatrix} -1 \\ \varphi \end{pmatrix}.$$

Both U and V are rotations, by the angles 31.7° and 58.3°. So the shear is the same as: rotate by $-58.3°$, stretch the x-axis by φ and shrink the y-axis by $\frac{1}{\varphi}$, and rotate by 31.7° (Figure 8).

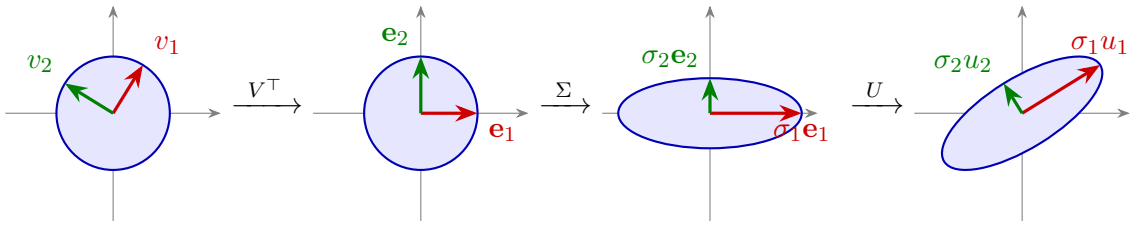


Figure 8: The SVD of the shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (Example 11.3). The rotation $V^\top$ moves the right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ to $\mathbf{e}_1, \mathbf{e}_2$; the scaling Σ stretches the axes by $\sigma_1 = \varphi$ and $\sigma_2 = 1/\varphi$; and the rotation U moves the axes to the left singular vectors. Altogether, the unit circle is mapped to an ellipse with semi-axes $\sigma_1\mathbf{u}_1 = A\mathbf{v}_1$ and $\sigma_2\mathbf{u}_2 = A\mathbf{v}_2$.

Proposition 11.4. *Let $A = U\Sigma V^\top$ be an SVD of a matrix A of rank r. Then*

$$A = \sigma_1\mathbf{u}_1\mathbf{v}_1^\top + \sigma_2\mathbf{u}_2\mathbf{v}_2^\top + \cdots + \sigma_r\mathbf{u}_r\mathbf{v}_r^\top,$$

and the four fundamental subspaces of A are

$$\begin{aligned} \operatorname{col}(A) &= \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}, & \operatorname{null}(A^\top) &= \operatorname{span}\{\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m\}, \\ \operatorname{row}(A) &= \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}, & \operatorname{null}(A) &= \operatorname{span}\{\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\}. \end{aligned}$$

Proof. For every $\mathbf{x}$, we saw in the proof of Theorem 11.1 that $A\mathbf{x} = \sum_{i \le r} \sigma_i \mathbf{u}_i(\mathbf{v}_i^\top \mathbf{x})$, which is the first formula applied to $\mathbf{x}$. The formula for $\operatorname{col}(A)$ was also shown there. If $\mathbf{x} = \sum_i c_i\mathbf{v}_i$, then $A\mathbf{x} = \sum_{i \le r} \sigma_i c_i \mathbf{u}_i$, which is $\mathbf{0}$ if and only if $c_1 = \cdots = c_r = 0$; this gives $\operatorname{null}(A)$. In an orthonormal basis, the orthogonal complement of the span of some basis vectors is the span of the other ones, so the remaining two formulas follow from Theorem 6.7 and Theorem 6.5(3). □

The first formula writes A as a sum of r matrices of rank 1. It also gives the reduced SVD $A = U_r\Sigma_r V_r^\top$, where U_r and V_r consist of the first r columns of U and V, and $\Sigma_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$; NumPy computes it with `np.linalg.svd(A, full_matrices=False)`.

11.3 Matrix norms and low-rank approximation

Definition 11.5 (Spectral norm)**.** The spectral norm (or operator norm) of an $m \times n$ matrix A is

$$\|A\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|},$$

the largest factor by which A stretches a vector.

Proposition 11.6. *Let A have singular values $\sigma_1 \ge \cdots \ge \sigma_r > 0$. Then*

$$\|A\|_2 = \sigma_1 \qquad \text{and} \qquad \|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}.$$

Proof. $\frac{\|A\mathbf{x}\|^2}{\|\mathbf{x}\|^2} = \frac{\mathbf{x}^\top A^\top A\mathbf{x}}{\mathbf{x}^\top\mathbf{x}}$ is the Rayleigh quotient of $A^\top A$, whose largest eigenvalue is σ_1^2; so the maximum is σ_1 by Proposition 10.7. For the Frobenius norm, $\|A\|_F^2 = \operatorname{tr}(A^\top A)$ is the sum of the eigenvalues of $A^\top A$ (by Lecture 1, the trace is the sum of the eigenvalues), which is $\sigma_1^2 + \cdots + \sigma_r^2$. □

The rank-1 expansion suggests approximating A by keeping only the terms with the largest singular values. This is the best possible approximation of a given rank.

Theorem 11.7 (Eckart–Young)**.** *Let A have the SVD $A = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$, and for $k < r$ let*

$$A_k = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^\top.$$

Then $\operatorname{rank}(A_k) = k$*, and for every matrix B of rank at most k,*

$$\|A - A_k\|_2 = \sigma_{k+1} \le \|A - B\|_2 \qquad \text{and} \qquad \|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2} \le \|A - B\|_F.$$

Proof. Since $A_k \mathbf{x} = \sum_{i \le k} \sigma_i (\mathbf{v}_i^\top \mathbf{x}) \mathbf{u}_i$ for every $\mathbf{x}$, and $A_k \mathbf{v}_i = \sigma_i \mathbf{u}_i$ for $i \le k$, the column space of A_k is $\operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$, so $\operatorname{rank}(A_k) = k$. Moreover, $A - A_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ is an SVD of $A - A_k$, with the singular values $\sigma_{k+1}, \ldots, \sigma_r$, so the formulas for $\|A - A_k\|_2$ and $\|A - A_k\|_F$ follow from Proposition 11.6. Now let $\operatorname{rank}(B) \le k$, and let A be $m \times n$.

Spectral norm. By the rank–nullity theorem, $\dim \operatorname{null}(B) \ge n - k$. The subspaces $\operatorname{null}(B)$ and $\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{k+1}\}$ of $\mathbb{R}^n$ have dimensions adding up to more than n, so they have a nonzero vector in common: together, a basis of the first and a basis of the second contain more than n vectors, so some nontrivial linear combination of them is 0, and moving the terms of the second basis to the other side gives a vector $\mathbf{x}$ that lies in both subspaces; it is nonzero, since each basis is linearly independent. We may assume $\|\mathbf{x}\| = 1$. Write $\mathbf{x} = \sum_{i \le k+1} c_i \mathbf{v}_i$ with $\sum_i c_i^2 = 1$. Then $B\mathbf{x} = \mathbf{0}$ and $A\mathbf{x} = \sum_{i \le k+1} \sigma_i c_i \mathbf{u}_i$, so

$$\|A - B\|_2^2 \ge \|(A - B)\mathbf{x}\|^2 = \|A\mathbf{x}\|^2 = \sum_{i=1}^{k+1} \sigma_i^2 c_i^2 \ge \sigma_{k+1}^2 \sum_{i=1}^{k+1} c_i^2 = \sigma_{k+1}^2.$$

Frobenius norm. Choose an orthonormal basis of $\operatorname{col}(B)$, which has at most k vectors, and add vectors of an orthonormal basis of $\operatorname{col}(B)^\perp$ (Theorem 6.5) until there are k vectors; this is possible since $k < r \le m$. Let W be their span, and C the $m \times k$ matrix with these columns, so that $C^\top C = I_k$, and $P = CC^\top$ is the matrix of the orthogonal projection onto $W \supseteq \operatorname{col}(B)$ (Proposition 8.7). For each column $\mathbf{a}_j$ of A and $\mathbf{b}_j$ of B, we have $\mathbf{a}_j - \mathbf{b}_j = (P\mathbf{a}_j - \mathbf{b}_j) + (\mathbf{a}_j - P\mathbf{a}_j)$ with $P\mathbf{a}_j - \mathbf{b}_j \in W$ and $\mathbf{a}_j - P\mathbf{a}_j \in W^\perp$, so by Pythagoras,

$$\|\mathbf{a}_j - \mathbf{b}_j\|^2 \ge \|\mathbf{a}_j - P\mathbf{a}_j\|^2 = \|\mathbf{a}_j\|^2 - \|P\mathbf{a}_j\|^2 = \|\mathbf{a}_j\|^2 - \|C^\top \mathbf{a}_j\|^2.$$

Adding over j gives $\|A - B\|_F^2 \ge \|A\|_F^2 - \|C^\top A\|_F^2 = \|A\|_F^2 - \operatorname{tr}(C^\top A A^\top C)$. The symmetric matrix $AA^\top = U\Sigma\Sigma^\top U^\top$ has the eigenvalues $\sigma_1^2 \ge \cdots \ge \sigma_r^2$, followed by zeros, so by Proposition 10.8, $\operatorname{tr}(C^\top A A^\top C) \le \sigma_1^2 + \cdots + \sigma_k^2$. Since $\|A\|_F^2 = \sigma_1^2 + \cdots + \sigma_r^2$, we get $\|A - B\|_F^2 \ge \sigma_{k+1}^2 + \cdots + \sigma_r^2$. □

Example 11.8. Let $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$. Then $A^\top A = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$ has the eigenvalues 45 and 5, with eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)^\top$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)^\top$. So $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$, and

$$\mathbf{u}_1 = \frac{1}{3\sqrt{5}} A\mathbf{v}_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad \mathbf{u}_2 = \frac{1}{\sqrt{5}} A\mathbf{v}_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

The best rank-1 approximation of A is

$$A_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top = 3\sqrt{5} \cdot \frac{1}{\sqrt{20}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{pmatrix},$$

with error $\|A - A_1\|_F = \|A - A_1\|_2 = \sigma_2 = \sqrt{5}$. Also, $\|A\|_2 = 3\sqrt{5} \approx 6.71$.

Remark 11.9 (Low-rank approximation in machine learning). A grayscale image with $m \times n$ pixels is an $m \times n$ matrix. Storing its rank-k approximation A_k requires only the $k(m+n+1)$ numbers σ_i, $\mathbf{u}_i$, $\mathbf{v}_i$, instead of mn; for natural images, the singular values decay quickly, and a small k already gives a good approximation (Programming Exercise P10). The same idea is behind recommender systems, which approximate the matrix of user ratings by a low-rank matrix, latent semantic analysis of text, and the compression of large neural networks.

11.4 The pseudo-inverse

Definition 11.10 (Pseudo-inverse)**.** Let $A = U\Sigma V^\top$ be an SVD of an $m \times n$ matrix A. The pseudo-inverse of A is the $n \times m$ matrix

$$A^+ = V\Sigma^+ U^\top,$$

where Σ^+ is the $n \times m$ matrix with $(\Sigma^+)_{ii} = \frac{1}{\sigma_i}$ for $i \le r$ and all other entries 0.

If A is invertible, then $A^+ = A^{-1}$, since $A^{-1} = (U\Sigma V^\top)^{-1} = V\Sigma^{-1}U^\top$.

Proposition 11.11. *If A has linearly independent columns, then $A^+ = (A^\top A)^{-1}A^\top$, and $A^+\mathbf{y}$ is the least squares solution of $A\mathbf{w} \approx \mathbf{y}$ (Theorem 8.10).*

Proof. In this case, $r = n$, and $A^\top A = V\Sigma^\top U^\top U\Sigma V^\top = V \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2)V^\top$. Hence

$$(A^\top A)^{-1}A^\top = V \operatorname{diag}(\sigma_1^{-2}, \dots, \sigma_n^{-2})V^\top V\Sigma^\top U^\top = V \operatorname{diag}(\sigma_1^{-2}, \dots, \sigma_n^{-2})\Sigma^\top U^\top = V\Sigma^+ U^\top.$$

The second statement is Theorem 8.10. □

In general, when the columns of A are dependent, the least squares problem has infinitely many solutions, and $A^+\mathbf{y}$ is the one with the smallest norm.

Proposition 11.12. *Let A be an $m \times n$ matrix and $\mathbf{y} \in \mathbb{R}^m$. The vectors $\mathbf{w}$ that minimize $\|A\mathbf{w} - \mathbf{y}\|$ are exactly the vectors $\mathbf{w} = A^+\mathbf{y} + \mathbf{v}$ with $\mathbf{v} \in \operatorname{null}(A)$, and among them, $A^+\mathbf{y}$ is the only one with the smallest norm.*

Proof. Let $A = U\Sigma V^\top$, and write $\mathbf{z} = V^\top\mathbf{w}$ and $\mathbf{c} = U^\top\mathbf{y}$. Since $U^\top$ preserves lengths (Theorem 4.7) and $U^\top A = \Sigma V^\top$,

$$\|A\mathbf{w} - \mathbf{y}\|^2 = \|U^\top(A\mathbf{w} - \mathbf{y})\|^2 = \|\Sigma\mathbf{z} - \mathbf{c}\|^2 = \sum_{i=1}^{r}(\sigma_i z_i - c_i)^2 + \sum_{i=r+1}^{m} c_i^2.$$

This is smallest if and only if $z_i = c_i/\sigma_i$ for $i \le r$, with $z_{r+1}, \dots, z_n$ arbitrary. For $z_{r+1} = \cdots = z_n = 0$, this is $\mathbf{z} = \Sigma^+\mathbf{c}$, i.e., $\mathbf{w} = V\Sigma^+U^\top\mathbf{y} = A^+\mathbf{y}$. In general, $\mathbf{w} = V\mathbf{z} = A^+\mathbf{y} + \sum_{i>r} z_i\mathbf{v}_i$, and the vectors $\sum_{i>r} z_i\mathbf{v}_i$ are exactly the elements of $\operatorname{null}(A)$ (Proposition 11.4). Finally, $\|\mathbf{w}\|^2 = \|\mathbf{z}\|^2 = \sum_{i\le r}(c_i/\sigma_i)^2 + \sum_{i>r} z_i^2$ is smallest exactly when $z_{r+1} = \cdots = z_n = 0$. □

NumPy's `np.linalg.pinv` and `np.linalg.lstsq` use the SVD in this way.

11.5 The condition number

Data are never exact, and computers round every number to about 16 significant digits. How much can such small errors change the solution of a linear system? The singular values give the answer.

Definition 11.13 (Condition number)**.** Let A be an $m \times n$ matrix with linearly independent columns, and singular values $\sigma_1 \geq \cdots \geq \sigma_n > 0$. Its condition number is $\kappa(A) = \frac{\sigma_1}{\sigma_n} \geq 1$.

For an invertible $n \times n$ matrix, $A^{-1} = V\Sigma^{-1}U^\top$ has the singular values $\frac{1}{\sigma_i}$, so $\kappa(A) = \|A\|_2\|A^{-1}\|_2$ by Proposition 11.6. For a symmetric matrix $A = Q\Lambda Q^\top$, the singular values are the numbers $|\lambda_i|$ (as $A^\top A = Q\Lambda^2 Q^\top$), so $\kappa(A) = \frac{\max_i |\lambda_i|}{\min_i |\lambda_i|}$. Orthogonal matrices have $\kappa(Q) = 1$, the best possible value.

Theorem 11.14 (Sensitivity of linear systems)**.** *Let A be invertible, $A\mathbf{x} = \mathbf{b} \neq \mathbf{0}$, and $A(\mathbf{x}+\boldsymbol{\delta}\mathbf{x}) = \mathbf{b} + \boldsymbol{\delta}\mathbf{b}$. Then*

$$\frac{\|\boldsymbol{\delta}\mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(A)\,\frac{\|\boldsymbol{\delta}\mathbf{b}\|}{\|\mathbf{b}\|},$$

with equality for $\mathbf{b} = \mathbf{u}_1$ and $\boldsymbol{\delta}\mathbf{b} = t\mathbf{u}_n$.

Proof. $\boldsymbol{\delta}\mathbf{x} = A^{-1}\boldsymbol{\delta}\mathbf{b}$, so $\|\boldsymbol{\delta}\mathbf{x}\| \leq \|A^{-1}\|_2\|\boldsymbol{\delta}\mathbf{b}\| = \frac{1}{\sigma_n}\|\boldsymbol{\delta}\mathbf{b}\|$, and $\|\mathbf{b}\| = \|A\mathbf{x}\| \leq \sigma_1\|\mathbf{x}\|$. Dividing the first inequality by $\|\mathbf{x}\| \geq \frac{1}{\sigma_1}\|\mathbf{b}\|$ gives the claim. For $\mathbf{b} = \mathbf{u}_1$, $\mathbf{x} = \frac{1}{\sigma_1}\mathbf{v}_1$, and for $\boldsymbol{\delta}\mathbf{b} = t\mathbf{u}_n$, $\boldsymbol{\delta}\mathbf{x} = \frac{t}{\sigma_n}\mathbf{v}_n$, so both inequalities are equalities. □

A relative error in $\mathbf{b}$ can thus be amplified by the factor $\kappa(A)$ in the solution. As a rule of thumb, solving a system with a condition number of 10^k in floating-point arithmetic loses about k of the 16 significant digits.

Example 11.15. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 1.001 \end{pmatrix}$, whose columns are almost parallel. Then $\kappa(A) \approx 4002$. The system $A\mathbf{x} = (2, 2.001)^\top$ has the solution $\mathbf{x} = (1, 1)^\top$, but changing the right side by 0.001 to $(2, 2.002)^\top$ gives $\mathbf{x} = (0, 2)^\top$: a relative change of 0.035% in $\mathbf{b}$ changes $\mathbf{x}$ by 100%.

The condition number explains how least squares problems should be solved. The singular values of $A^\top A$ are σ_i^2, so

$$\kappa(A^\top A) = \kappa(A)^2.$$

Solving the normal equations $A^\top A\mathbf{w} = A^\top\mathbf{y}$ squares the condition number, and may lose twice as many digits as necessary. With a QR decomposition $A = QR$ (Proposition 5.9), the normal equations become $R^\top R\mathbf{w} = R^\top Q^\top\mathbf{y}$, i.e., $R\mathbf{w} = Q^\top\mathbf{y}$, and $\kappa(R) = \kappa(A)$, since multiplying by Q does not change lengths. This is why `np.linalg.lstsq` uses the SVD and never forms $A^\top A$. Regularization also helps: ridge regression solves $(A^\top A + \lambda I)\mathbf{w} = A^\top\mathbf{y}$, whose matrix has the condition number $\frac{\sigma_1^2+\lambda}{\sigma_n^2+\lambda} \leq \kappa(A)^2$. In Lecture 3, the condition number of the Hessian determines how fast gradient descent converges.

11.6 Principal component analysis

Principal component analysis (PCA) is the most widely used method to reduce the dimension of data. Let $\mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^d$ be data points, which we assume to be centered, i.e., $\sum_i \mathbf{x}_i = \mathbf{0}$ (otherwise, subtract the mean from every data point first). Let X be the $N \times d$ data matrix whose rows are $\mathbf{x}_1^\top, \ldots, \mathbf{x}_N^\top$, and let

$$S = \frac{1}{N}X^\top X = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i\mathbf{x}_i^\top$$

be the (sample) covariance matrix, which is symmetric positive semidefinite.

We look for a k-dimensional subspace U such that the projected data points $\pi_U(\mathbf{x}_i)$ are as close as possible to the original ones, i.e., such that the reconstruction error $\sum_i \|\mathbf{x}_i - \pi_U(\mathbf{x}_i)\|^2$ is as small as possible. By Pythagoras, $\|\mathbf{x}_i\|^2 = \|\pi_U(\mathbf{x}_i)\|^2 + \|\mathbf{x}_i - \pi_U(\mathbf{x}_i)\|^2$, so equivalently, we look for the subspace for which the projected data has the largest spread $\sum_i \|\pi_U(\mathbf{x}_i)\|^2$.

Theorem 11.16 (PCA)**.** *Let $\mathbf{q}_1, \dots, \mathbf{q}_d$ be orthonormal eigenvectors of S with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$. Among all k-dimensional subspaces U of $\mathbb{R}^d$, the reconstruction error is smallest for $U = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$, and then*

$$\frac{1}{N}\sum_{i=1}^{N} \|\mathbf{x}_i - \pi_U(\mathbf{x}_i)\|^2 = \lambda_{k+1} + \cdots + \lambda_d.$$

Proof. Let $\mathbf{b}_1, \dots, \mathbf{b}_k$ be an orthonormal basis of U. Then $\pi_U(\mathbf{x}_i) = \sum_j (\mathbf{b}_j^\top \mathbf{x}_i)\mathbf{b}_j$, so $\|\pi_U(\mathbf{x}_i)\|^2 = \sum_j (\mathbf{b}_j^\top \mathbf{x}_i)^2$ by Pythagoras, and

$$\frac{1}{N}\sum_{i=1}^{N} \|\pi_U(\mathbf{x}_i)\|^2 = \sum_{j=1}^{k} \frac{1}{N}\sum_{i=1}^{N} (\mathbf{b}_j^\top \mathbf{x}_i)^2 = \sum_{j=1}^{k} \frac{1}{N}\|X\mathbf{b}_j\|^2 = \sum_{j=1}^{k} \mathbf{b}_j^\top S \mathbf{b}_j,$$

which by Proposition 10.8 is at most $\lambda_1 + \cdots + \lambda_k$, with equality for $\mathbf{b}_j = \mathbf{q}_j$. For $k = 1$, this is just Proposition 10.7. The error formula follows from Pythagoras, since $\frac{1}{N}\sum_i \|\mathbf{x}_i\|^2 = \frac{1}{N}\operatorname{tr}(X^\top X) = \operatorname{tr}(S) = \lambda_1 + \cdots + \lambda_d$. □

The vectors $\mathbf{q}_1, \mathbf{q}_2, \dots$ are called the principal components, or principal directions, of the data, and $\lambda_i = \mathbf{q}_i^\top S \mathbf{q}_i$ is the variance of the data in the direction $\mathbf{q}_i$. The fraction $\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d}$ is the proportion of the variance explained by the first k principal components. In practice, PCA is computed with the SVD of the data matrix: if $X = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$, then

$$S = \frac{1}{N} X^\top X = \tilde{V}\,\frac{\tilde{\Sigma}^\top \tilde{\Sigma}}{N}\,\tilde{V}^\top,$$

so the principal components are the right singular vectors of X, and $\lambda_i = \sigma_i^2/N$. The coordinates of the data in the principal directions are the entries of $X\tilde{V}$.

Example 11.17. Consider the four centered data points $(2,2)$, $(-2,-2)$, $(1,-1)$ and $(-1,1)$ in $\mathbb{R}^2$. Then

$$X^\top X = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}, \qquad S = \frac{1}{4} X^\top X = \begin{pmatrix} 2.5 & 1.5 \\ 1.5 & 2.5 \end{pmatrix},$$

with eigenvalues 4 and 1 and eigenvectors $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$ and $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. The first principal component explains $\frac{4}{4+1} = 80\%$ of the variance. Projecting onto $\operatorname{span}\{\mathbf{q}_1\}$ keeps the first two points, and maps the last two to $\mathbf{0}$; the average reconstruction error is $\frac{1}{4}(0+0+2+2) = 1 = \lambda_2$, as predicted by Theorem 11.16.

12 Key Takeaways

- A norm measures length: it is positive definite, absolutely homogeneous, and satisfies the triangle inequality. Important examples on $\mathbb{R}^n$ are the ℓ_1, ℓ_2 and ℓ_∞ norms, whose unit balls are a diamond, a ball and a cube.

- An inner product is a symmetric, positive definite, bilinear function. On an n-dimensional space with a fixed basis, inner products correspond to symmetric positive definite matrices: $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top A \mathbf{y}$.
- Every inner product induces a norm $\|x\| = \sqrt{\langle x, x \rangle}$ and a distance $d(x, y) = \|x - y\|$. The Cauchy–Schwarz inequality $|\langle x, y \rangle| \leq \|x\| \|y\|$ gives the triangle inequality and allows us to define angles by $\cos \omega = \frac{\langle x, y \rangle}{\|x\| \|y\|}$.
- Vectors are orthogonal if their inner product is 0. Orthogonal matrices ($Q^\top Q = I$) are exactly the matrices that preserve dot products, lengths and angles.
- In an orthonormal basis, coordinates are inner products: $v = \sum_i \langle v, b_i \rangle b_i$. The Gram–Schmidt process turns any basis into an orthonormal basis, and gives the QR decomposition.
- Every subspace U of a finite-dimensional inner product space has an orthogonal complement with $V = U \oplus U^\perp$. For a matrix, $\operatorname{null}(A) = \operatorname{row}(A)^\perp$ and $\operatorname{null}(A^\top) = \operatorname{col}(A)^\perp$.
- A hyperplane $\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} = c\}$ has normal vector $\mathbf{w}$; linear classifiers use hyperplanes as decision boundaries, and the distance of $\mathbf{x}$ to the hyperplane is $\frac{|\mathbf{w}^\top \mathbf{x} - c|}{\|\mathbf{w}\|}$.
- The orthogonal projection onto $U = \operatorname{col}(B)$ is $\pi_U(\mathbf{x}) = B(B^\top B)^{-1} B^\top \mathbf{x}$; it is the point of U closest to $\mathbf{x}$. Least squares regression projects the target vector onto the column space of the design matrix, which leads to the normal equations $A^\top A \mathbf{w} = A^\top \mathbf{y}$.
- Rotations are orthogonal matrices with determinant 1. They preserve lengths, angles and orientation; rotations of the plane commute, but rotations of $\mathbb{R}^3$ do not.
- Spectral theorem: a symmetric matrix has real eigenvalues and an orthonormal basis of eigenvectors, $A = Q \Lambda Q^\top = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$. It is positive definite if and only if all eigenvalues are positive, and $\max_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x} = \lambda_{\max}$; more generally, λ_k is the max-min of the Rayleigh quotient over k-dimensional subspaces (Courant–Fischer), so eigenvalues of symmetric matrices move by at most ε under perturbations with eigenvalues in $[-\varepsilon, \varepsilon]$ (Weyl).
- A symmetric matrix is positive definite if and only if $A = LL^\top$ with a lower triangular L with positive diagonal (Cholesky), the standard way to solve systems with such matrices and to compute $\det(A)$. Positive semidefinite matrices have a unique positive semidefinite square root $A^{1/2} = Q \Lambda^{1/2} Q^\top$.
- The Laplacian $L = D - W$ of a weighted graph satisfies $\mathbf{x}^\top L \mathbf{x} = \sum_{i<j} w_{ij} (x_i - x_j)^2$, so it is positive semidefinite. The multiplicity of its eigenvalue 0 is the number of connected components, and its second smallest eigenvalue λ_2 is a lower bound for the ratio cut of every split of the vertices. Spectral clustering splits the vertices by the signs of the Fiedler vector $\mathbf{q}_2$.
- Every matrix has a singular value decomposition $A = U \Sigma V^\top = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$: rotate, scale the axes by the singular values, rotate. The σ_i^2 are the eigenvalues of $A^\top A$, $r = \operatorname{rank}(A)$, $\|A\|_2 = \sigma_1$ and $\|A\|_F^2 = \sum_i \sigma_i^2$.
- The condition number $\kappa(A) = \sigma_1 / \sigma_n$ bounds the amplification of relative errors in linear systems, $\frac{\|\boldsymbol{\delta}\mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(A) \frac{\|\boldsymbol{\delta}\mathbf{b}\|}{\|\mathbf{b}\|}$. Since $\kappa(A^\top A) = \kappa(A)^2$, least squares problems are solved with QR or the SVD rather than the normal equations.

- Keeping the k largest singular values gives the best rank-k approximation (Eckart–Young), with applications to compression and recommender systems. PCA projects centered data onto the eigenvectors of the covariance matrix with the largest eigenvalues, i.e., onto the top right singular vectors of the data matrix.

13 Exercises

1. Let $\mathbf{x} = (1, -2, 2)^\top$ and $\mathbf{y} = (3, 0, 1)^\top$. Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$, and the distance between $\mathbf{x}$ and $\mathbf{y}$ in each of the three norms.

2. Show that $\|\mathbf{x}\|_2 \le \sqrt{n}\,\|\mathbf{x}\|_\infty$ for all $\mathbf{x} \in \mathbb{R}^n$. For which vectors $\mathbf{x}$ does equality hold in $\|\mathbf{x}\|_1 \le \sqrt{n}\,\|\mathbf{x}\|_2$?

3. Show that the ℓ_∞ norm on $\mathbb{R}^2$ does not satisfy the parallelogram law, and conclude that it does not come from an inner product.

4. Which of the following matrices are symmetric positive definite? For each one that is, write down the corresponding inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top A \mathbf{y}$ on $\mathbb{R}^2$ explicitly.

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}.$$

5. Find the angle between the vectors: (a) $(1, 1, 0)^\top$ and $(1, 0, 1)^\top$; (b) $(1, 2, 3)^\top$ and $(-3, 0, 1)^\top$; (c) $(1, 0)^\top$ and $(-1, \sqrt{3})^\top$.

6. (a) Find all vectors in $\mathbb{R}^3$ that are orthogonal to both $(1, 1, 1)^\top$ and $(1, -1, 1)^\top$. (Hint: this is the null space of a matrix.)

   (b) The planes $x + y + z = 3$ and $x - y + z = 5$ intersect in a line ℓ. Find an equation of the plane through the origin that is perpendicular to ℓ.

7. Apply the Gram–Schmidt process to $\mathbf{v}_1 = (1, 2, 2)^\top$ and $\mathbf{v}_2 = (1, 0, 1)^\top$. Then find a third vector that extends the result to an orthonormal basis of $\mathbb{R}^3$, and write $(1, 1, 1)^\top$ in this basis. Check Parseval's identity.

8. Find a basis of the orthogonal complement of $U = \operatorname{span}\{(1, 0, 1, 0)^\top, (0, 1, 0, 1)^\top\}$ in $\mathbb{R}^4$.

9. Let $\mathbf{x} = (1, 0, 2)^\top$. Compute the orthogonal projection of $\mathbf{x}$ onto (a) the line spanned by $(1, 1, 0)^\top$, and (b) the plane $x_1 + x_2 + x_3 = 0$. Write down the projection matrices of both projections.

10. Find the point of the plane $2x + y + 3z = 3$ that is closest to the origin, and the distance from the origin to this plane.

11. Let P be a symmetric $n \times n$ matrix with $P^2 = P$. Show that P is the matrix of the orthogonal projection onto $\operatorname{col}(P)$. (Hint: show that $\mathbf{x} - P\mathbf{x} \in \operatorname{col}(P)^\perp = \operatorname{null}(P^\top)$.)

12. Let $\mathbf{u} \in \mathbb{R}^n$ be a unit vector, and let $H = I_n - 2\mathbf{u}\mathbf{u}^\top$.

    (a) Show that H is symmetric and orthogonal, and that $H^2 = I_n$.

    (b) Show that $H\mathbf{u} = -\mathbf{u}$, and that $H\mathbf{x} = \mathbf{x}$ for every $\mathbf{x} \perp \mathbf{u}$. Describe H geometrically. (Such matrices are called Householder reflections; `np.linalg.qr` uses them.)

13. Find the polynomial p of degree at most 1 that minimizes $\int_{-1}^{1} \big(x^3 - p(x)\big)^2\,dx$, by projecting x^3 onto $\operatorname{span}\{1, x\}$ in $C[-1,1]$. Compare with Example 7.3.

14. Find the least squares line for the data points $(0,1)$, $(1,1)$ and $(2,3)$, and check that the residual vector is orthogonal to the columns of the design matrix.

15. Show that the product of two orthogonal 2×2 matrices with determinant -1 (for example, two reflections) is a rotation. Compute $R_1(\theta)^\top R_1(\theta)$ and $\det R_1(\theta)$.

16. Use the Cauchy–Schwarz inequality to show that $(x_1 + \cdots + x_n)^2 \le n(x_1^2 + \cdots + x_n^2)$. Conclude that if $a, b, c \ge 0$ and $a + b + c = 1$, then $a^2 + b^2 + c^2 \ge \frac{1}{3}$. When does equality hold?

17. Find the spectral decomposition $A = Q\Lambda Q^\top$ of $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$, and write A as $\lambda_1 \mathbf{q}_1\mathbf{q}_1^\top + \lambda_2 \mathbf{q}_2\mathbf{q}_2^\top$. Is A positive definite? Conversely, show that every matrix of the form $Q\Lambda Q^\top$, with Q orthogonal and Λ diagonal, is symmetric.

18. Find the largest and the smallest value of $\mathbf{x}^\top A\mathbf{x}$ over all unit vectors $\mathbf{x} \in \mathbb{R}^2$, for $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and the unit vectors where they are attained.

19. (Cholesky decomposition and square roots.)

    (a) Find the Cholesky decomposition of $\begin{pmatrix} 4 & 2 \\ 2 & 10 \end{pmatrix}$. What goes wrong when you apply the algorithm to $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, and why?

    (b) Compute the square root of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

    (c) Let A be symmetric positive definite with the Cholesky decomposition $A = LL^\top$. Show that $Q = L^{-1}A^{1/2}$ is orthogonal, so that $A^{1/2} = LQ$. More generally, show that if $A = BB^\top = CC^\top$ with invertible matrices B and C, then $B^{-1}C$ is orthogonal.

20. (Courant–Fischer.) Let A be a symmetric $n \times n$ matrix with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n$, and let A' be its upper left $(n-1)\times(n-1)$ submatrix, with eigenvalues $\mu_1 \ge \cdots \ge \mu_{n-1}$.

    (a) Show that $\lambda_k \ge \mu_k$ for $k \le n-1$. (Hint: the subspaces of $\mathbb{R}^{n-1}$ are subspaces of $\mathbb{R}^n$ after appending a zero coordinate.)

    (b) Apply (a) to $-A$ to show that $\mu_k \ge \lambda_{k+1}$. So the eigenvalues of A' interlace with those of A (Cauchy's interlacing theorem).

    (c) Check the interlacing for the matrix $J + 2I_3$ of Example 10.2.

21. Let L be the Laplacian of a graph with weights w_{ij}.

    (a) Show that $L = \sum_{i<j} w_{ij}(\mathbf{e}_i - \mathbf{e}_j)(\mathbf{e}_i - \mathbf{e}_j)^\top$, and use this to give another proof of Proposition 10.13.

    (b) Find the Laplacian and its eigenvalues for the star graph with the center 1 and the leaves $2, \ldots, n$, where $n \ge 3$ and all weights are 1. (Hint: consider the vectors with $x_1 = 0$ whose entries add up to 0, and the vector $(n-1, -1, \ldots, -1)^\top$.) Why is the Fiedler vector not unique, and what does this say about clusters in a star?

22. Compute an SVD of (a) $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$ and (b) $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$. For (b), write A as $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$, and compute the pseudo-inverse A^+.

23. Show that:

    (a) all singular values of an orthogonal matrix are equal to 1;

    (b) the singular values of a symmetric matrix are the absolute values of its nonzero eigenvalues;

    (c) $A^\top A$ and $AA^\top$ have the same nonzero eigenvalues;

    (d) $\|A\|_2 = \|A^\top\|_2$.

24. (Condition numbers.)

    (a) Show that $\kappa(cA) = \kappa(A)$ for $c \neq 0$, that $\kappa(QA) = \kappa(A)$ for an orthogonal matrix Q, and that $\kappa(A^{-1}) = \kappa(A)$ for an invertible matrix A.

    (b) Compute the condition numbers of $\begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix}$ for $0 < \varepsilon < 1$, and of the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

    (c) Show that $\kappa(A^\top A + \lambda I) = \frac{\sigma_1^2 + \lambda}{\sigma_n^2 + \lambda}$ for $\lambda > 0$, and that this is a decreasing function of λ if $\sigma_1 > \sigma_n$.

25. Carry out PCA by hand for the four data points $(3, 3)$, $(-3, -3)$, $(1, -1)$, $(-1, 1)$: compute the covariance matrix, the principal components, and the proportion of the variance explained by the first principal component. Project the data onto the first principal component, and compute the average reconstruction error.

14 Programming Exercises

As in Lecture 1, implement the algorithms yourself with NumPy, and use NumPy's built-in routines only to check your results. You need Matplotlib for P1, P5, P7, P10 and P11. Start each file with

```
import numpy as np
rng = np.random.default_rng(0)  # random number generator for test data
```

P1. Norms and unit balls. Write a function `p_norm(x, p)` for $p \geq 1$ and $p = \infty$ (use `np.inf`), and compare it with `np.linalg.norm(x, ord=p)`.

- For 10 000 random vectors in $\mathbb{R}^5$, check the inequalities $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{5}\,\|\mathbf{x}\|_2$.
- Plot the unit circles $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_p = 1\}$ for $p = 1, 1.5, 2, 4, \infty$ in one picture, as in Figure 1. (Hint: for each angle t, rescale the vector $(\cos t, \sin t)$ to have p-norm 1.)

P2. Inner products and angles. Write a function `inner(x, y, A)` that computes $\mathbf{x}^\top A \mathbf{y}$, and functions `norm_A(x, A)` and `angle_A(x, y, A)` for the induced norm and angle.

- Check Example 4.4: the angle between $(1, 1)$ and $(-1, 1)$ is $90°$ for $A = I_2$, but about $109.5°$ for $A = \mathrm{diag}(2, 1)$.
- Write a function `is_spd(A)` that tests whether a symmetric 2×2 matrix is positive definite, using Proposition 2.7. For larger matrices, NumPy's `np.linalg.cholesky(A)` succeeds exactly for symmetric positive definite matrices (Theorem 2.10) and raises `np.linalg.LinAlgError` otherwise; use it to test the matrices of Exercise 4.

- For random vectors, check the Cauchy–Schwarz inequality, and the parallelogram law for the ℓ_2 norm (but not for the ℓ_1 norm).

P3. Gram–Schmidt, and how it can fail. Implement the classical Gram–Schmidt process `gram_schmidt(V)`, which returns a matrix Q whose columns are the orthonormal vectors obtained from the columns of `V`, exactly as in Theorem 5.6.

- Check your function on Example 5.7, and compute $R = Q^\top V$; it should be upper triangular, with $V = QR$.
- The modified Gram–Schmidt process subtracts the components one at a time, from the updated vector: for each $i < j$, replace $\mathbf{u}_j$ by $\mathbf{u}_j - \langle \mathbf{u}_j, \mathbf{e}_i \rangle \mathbf{e}_i$. In exact arithmetic, both versions give the same result. Implement it as well.
- For the Hilbert matrices H_n with entries $\frac{1}{i+j-1}$ (whose columns are nearly linearly dependent) and $n = 6, 8, 10, 12$, compute the loss of orthogonality `np.linalg.norm(Q.T @ Q - np.eye(n))` for the classical version, the modified version, and `np.linalg.qr`. What do you observe?

P4. Projections. Write a function `project(x, B)` that returns the orthogonal projection of $\mathbf{x}$ onto the column space of `B`, by solving the normal equations, and a function `projection_matrix(B)`.

- Check your functions on Example 8.6.
- For a random 6×3 matrix B and a random $\mathbf{x} \in \mathbb{R}^6$, check that $P^2 = P$, $P^\top = P$, and that $\mathbf{x} - P\mathbf{x}$ is orthogonal to every column of B.
- Check Theorem 8.8 numerically: compare $\|\mathbf{x} - P\mathbf{x}\|$ with $\|\mathbf{x} - B\boldsymbol{\lambda}\|$ for 1000 random vectors $\boldsymbol{\lambda} \in \mathbb{R}^3$.
- Orthonormalize the columns of B with your function from P3, and check that $QQ^\top$ is the same projection matrix.

P5. Polynomial regression. Generate 20 noisy data points

```
x = np.linspace(-1, 1, 20)
y = np.sin(3 * x) + rng.normal(0, 0.1, size=20)
```

- For $d = 1, 3, 5, 15$, fit a polynomial of degree d by least squares: build the design matrix with columns $1, x, \ldots, x^d$ and solve the normal equations. Compare your coefficients with `np.linalg.lstsq`.
- Plot the data and the four fitted polynomials. Which degree fits the training data best? Which one would you trust for new data? (This is the phenomenon of overfitting.)
- Check that for each d, the residual vector is orthogonal to all columns of the design matrix.

P6. Nearest neighbors with different metrics. Generate two classes of points in $\mathbb{R}^2$:

```
X0 = rng.normal([0, 0], [1.0, 0.3], size=(100, 2))   # class 0
X1 = rng.normal([1.5, 1.0], [1.0, 0.3], size=(100, 2))  # class 1
```

and split the 200 points randomly into 150 training points and 50 test points.

- Implement the 1-nearest neighbor classifier with the ℓ_1, ℓ_2 and ℓ_∞ distances, and with the cosine similarity (largest similarity instead of smallest distance). Compare their accuracies on the test points.
- Multiply the second coordinate of all points by 1000 (as if it were measured in different units) and repeat. What happens to the accuracies of the distance-based and the cosine-based classifiers, and why? What happens if you standardize each feature (subtract its mean, divide by its standard deviation) first?
- Rotate all points by a random rotation R_θ and repeat. Which classifiers give exactly the same predictions, and why?

P7. Rotations. Write functions `R1(t)`, `R2(t)` and `R3(t)` that return the rotation matrices of Section 9, and a function `givens(n, i, j, t)`.

- Check that all of them are orthogonal with determinant 1, and that they preserve the lengths of random vectors.
- Check that `R1(t) @ R3(s)` and `R3(s) @ R1(t)` differ for most angles, while `R3(t) @ R3(s)` equals `R3(t + s)`.
- Let `Q, R = np.linalg.qr(rng.normal(size=(3, 3)))`. Then `Q` is orthogonal, but its determinant may be -1. How can you turn it into a rotation by changing the sign of one column? Generate many random rotations this way, apply them to $\mathbf{e}_3$, and plot the resulting points on the sphere (Matplotlib can make 3D scatter plots).

P8. The perceptron and its margin. The perceptron algorithm learns a linear classifier $f(\mathbf{x}) = \operatorname{sign}(\mathbf{w}^\top \mathbf{x} + b)$ from labeled data $(\mathbf{x}_i, y_i)$ with $y_i \in \{+1, -1\}$: start with $\mathbf{w} = \mathbf{0}$ and $b = 0$, and repeatedly go through the data; whenever a point is misclassified, i.e., $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \le 0$, update $\mathbf{w} \leftarrow \mathbf{w} + y_i \mathbf{x}_i$ and $b \leftarrow b + y_i$. Stop when all points are classified correctly.

- Generate linearly separable data, e.g., 50 points on each side of the line $\mathbf{x}_1 + \mathbf{x}_2 = 2$ (discard points that are closer than 0.2 to the line), and run the perceptron.
- Using Theorem 8.13, compute the margin of the learned classifier, i.e., the smallest distance of a training point to the decision boundary. Compare it with the margin of the classifier $\mathbf{w} = (1, 1)$, $b = -2$.
- Count the number of updates the perceptron needs. How does it change when you discard points closer than 0.5 (instead of 0.2) to the line?

P9. The spectral theorem in practice. For symmetric matrices, use `np.linalg.eigh`, which returns real eigenvalues in increasing order and orthonormal eigenvectors.

- For a random symmetric matrix `A = B + B.T` (with `B = rng.normal(size=(5, 5))`), check that `Q.T @ Q` is close to I_5 and that `Q @ np.diag(lam) @ Q.T` is close to A. Check the spectral decomposition $A = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$ as well.
- Write a function `is_spd(A)` based on Corollary 10.3, and compare it with your test from P2.
- Check Proposition 10.7 numerically: evaluate the Rayleigh quotient $\frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$ at 10 000 random vectors, and compare the largest and smallest values you find with the largest and smallest eigenvalues.

P10. Image compression with the SVD. Load a grayscale image as a 2D array `A`, for example with `A = plt.imread("photo.png").mean(axis=2)` for a color image (with `import matplotlib.pyplot as plt`), or create one, e.g., from the function $\sin(x)\cos(y) + \frac{x}{10}$ on a 200×300 grid plus some noise.

- Compute `U, s, Vt = np.linalg.svd(A, full_matrices=False)` and plot the singular values on a logarithmic scale. How fast do they decay?
- For $k = 1, 5, 20, 50$, compute the rank-k approximation A_k (Theorem 11.7) and display it as an image. Compare `np.linalg.norm(A - A_k)` with $\sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}$, and compute the storage ratio $\frac{k(m+n+1)}{mn}$.
- Check numerically that $\|A\|_2 = \sigma_1$ (`np.linalg.norm(A, 2)`) and $\|A\|_F^2 = \sum_i \sigma_i^2$.

P11. Principal component analysis. Generate 500 data points from a correlated Gaussian distribution:

```
C = np.array([[3.0, 1.5], [1.5, 1.0]])  # covariance matrix
X = rng.multivariate_normal([2, -1], C, size=500)
```

- Center the data, and compute the principal components in two ways: as eigenvectors of the covariance matrix $S = \frac{1}{N} X^\top X$ (with `np.linalg.eigh`), and as right singular vectors of X (with `np.linalg.svd`). Check that they agree (up to sign), and that $\lambda_i = \sigma_i^2 / N$. How do they compare with the eigenvectors of C?
- Plot the data together with the two principal directions, scaled by $\sqrt{\lambda_i}$. What proportion of the variance does the first principal component explain?
- Project the data onto the first principal component, plot the projected points, and check that the average reconstruction error is λ_2 (Theorem 11.16).
- Generate 500 points in $\mathbb{R}^{10}$ that lie close to a 2-dimensional subspace, for example

  ```
  X = rng.normal(size=(500, 2)) @ rng.normal(size=(2, 10))
  X = X + 0.05 * rng.normal(size=(500, 10))  # small noise
  ```

  Plot the 10 eigenvalues of the covariance matrix. How can you see from this plot that the data is essentially 2-dimensional?

P12. Spectral clustering. Generate 100 points on each of two concentric circles with the radii 1 and 3:

```
t = rng.uniform(0, 2 * np.pi, 200)
r = np.repeat([1.0, 3.0], 100)
X = np.c_[r * np.cos(t), r * np.sin(t)] + 0.1 * rng.normal(size=(200, 2))
```

- Build the weights $w_{ij} = e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2/(2s^2)}$ with $s = 0.5$ (and $w_{ii} = 0$), the Laplacian L (Definition 10.12), and its eigenvalues and eigenvectors with `np.linalg.eigh`. Check that the smallest eigenvalue is 0, with the eigenvector $\frac{1}{\sqrt{n}}\mathbf{1}$ (up to sign).
- Cluster the points by the signs of the Fiedler vector, and plot them in two colors. Does spectral clustering find the two circles? Why would a method that only splits the plane by a line fail here?

- Repeat with $s = 2$ and $s = 0.2$. How do the second and third smallest eigenvalues change, and how does the clustering change? Relate your observations to Theorem 10.14.

P13. Conditioning and least squares.

- Write a function `cholesky(A)` following the proof of Theorem 2.10, and compare it with `np.linalg.cholesky` on a random matrix $B^\top B + I$. Use it to solve $A\mathbf{x} = \mathbf{b}$ by forward and back substitution.
- The Hilbert matrix H_n has the entries $\frac{1}{i+j-1}$. For $n = 4, 6, \ldots, 14$, compute $\kappa(H_n)$ with `np.linalg.cond`, solve $H_n\mathbf{x} = H_n\mathbf{1}$, and plot the error $\|\mathbf{x} - \mathbf{1}\|$ against $\kappa(H_n)$ on logarithmic axes. How many correct digits does the rule of thumb after Theorem 11.14 predict?
- Fit a polynomial of degree 12 to 50 points $(t_i, \sin 4t_i)$ with t_i equally spaced in $[0, 1]$, using the columns $1, t, \ldots, t^{12}$ of a design matrix A. Compare $\kappa(A)$ with $\kappa(A^\top A)$, and compare the residuals of three solutions: the normal equations with `np.linalg.solve`, the QR decomposition, and `np.linalg.lstsq`.